Notes on Advanced Macroeconomics

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Preface

Aim of this work of mine is to provide clear macroeconomic knowledges in language as simple as possible. Most of this note is taken from the advanced macroeconomics course of Yanfei Deng I took. This note is still a work in progress and needs tons of ammending. See the newest version at `https://github.com/xolarvill/notes-on-economics` along with my other notes. And pull a request if you figure some part of this note is wrong.

This work was first written in markdown language on Obsidian, which provides a lightweight and highly personalized working experience. But it was later transfered into purely latex language on Sublime Text, due to poor support for rendering of long math equation blocks of Obsidian. For the tranfer I used Pandoc at `https://pandoc.org/`. I wrote some LaTeX snippets of Sublime Text to optimize the writing experience, and uploaded on github at `https://github.com/xolarvill/snippets-for-quick-latex-on-st`.

One can find enormous academical help on advanced macroeconomics from books listed below:

- Introduction to Modern Economic Growth, D. Acemoglu
- Economic Growth, R. Barro and S.i.M. Xavi
- The ABCs of RBCs, An Introduction to Dynamic Macroeconomic Models, G. McCandless
- Macroeconomics, A Comprehensive Textbook for First-Year Ph.D. Courses in Macroeconomics. M. Azzimonti, P. Krusell, A. McKay, and T. Mukoyama
- Adavanced Macroeconomics, D. Romer

Part I
Introduction

A short bridge from intermediate macroeconomics to the decisions, equilibrium concepts, and notation used in these notes.

1 Introduction

These notes assume intermediate macroeconomics, including national accounts, basic growth theory, and aggregate demand and supply. We now ask how aggregate outcomes follow from decisions over time, and how those decisions respond to the environment and policy.

Parts II–IV cover growth, fluctuations, and the open economy. Part V studies money, finance, and policy; Part VI applies these mechanisms to the dollar system and China's growth after the pandemic. Chapters and sections marked *planned* contain outlines for later development.

1.1 Growth, fluctuations, and their interaction

Growth theory asks how productive capacity and living standards evolve. Starting with capital accumulation, we introduce saving decisions, population, technology, and resource allocation to explain differences across economies.

Business-cycle theory asks how an economy responds to disturbances in productivity, demand, or financing. We identify the shock, then explain the response of spending, employment, and prices. A mechanism that amplifies a shock need not be its cause.

The questions overlap. A recession can reduce investment and change future capital. Population and productivity growth can change the conditions for stabilization policy. Growth need not be an unaffected trend, and fluctuations can have lasting effects. We begin with separate models to isolate mechanisms, then study their interaction.

1.2 From aggregate relations to individual decisions

An aggregate consumption function such as $C = C(Y - T)$ summarizes a relationship. It does not yet explain how a household responds to an expected tax change, uncertain income, or a borrowing limit. To study those questions, we specify what the household chooses and what constrains it.

A two-period example. Let a household receive real income y_0 today and y_1 tomorrow, with no initial or terminal assets. Saving a_1 earns a known gross real return $R > 0$; negative a_1 is borrowing at that return. The problem is

$$\begin{aligned} \max_{c_0, c_1, a_1} \; & u(c_0) + \beta u(c_1), \qquad 0 < \beta < 1, \\ \text{s.t. } & c_0 + a_1 = y_0, \\ & c_1 = y_1 + R a_1. \end{aligned} \tag{1}$$

Assume $u' > 0$, $u'' < 0$, positive consumption, and an interior solution. Substituting the constraints leaves one choice:

$$\max_{a_1} \; u(y_0 - a_1) + \beta u(y_1 + R a_1). \tag{2}$$

Differentiate with respect to a_1, then substitute back for consumption:

$$-u'(y_0 - a_1) + \beta R u'(y_1 + R a_1) = 0, \tag{3}$$

$$u'(c_0) = \beta R u'(c_1). \tag{4}$$

Saving one more unit costs $u'(c_0)$ units of utility today and yields $\beta R u'(c_1)$ tomorrow. An interior optimum equates the two.

For marginal utility $u'(c) = c^{-\sigma}$, $\sigma > 0$,

$$c_0^{-\sigma} = \beta R c_1^{-\sigma}, \tag{5}$$

$$\left(\frac{c_1}{c_0}\right)^{\sigma} = \beta R, \tag{6}$$

$$\frac{c_1}{c_0} = (\beta R)^{1/\sigma}. \tag{7}$$

Thus consumption growth depends on patience and the return to saving. Its sensitivity to that return is governed by $1/\sigma$, the elasticity of intertemporal substitution in this example. A binding borrowing constraint would require an inequality condition instead of this interior equality.

Expectations and policy. With uncertainty, today's utility cost is known but tomorrow's gain is not. Replace that gain by its conditional expectation. For an asset with gross real payoff R_{t+1}, the interior condition becomes

$$u'(c_t) = \beta E_t[R_{t+1} u'(c_{t+1})], \qquad E_t[\cdot] \equiv E[\cdot \mid \mathscr{I}_t]. \tag{8}$$

Here $\mathscr{I}_t$ is information available when the household chooses saving. An uncertain return stays inside the expectation; its comovement with marginal utility matters. Rational expectations require forecasts consistent with the model and available information, not knowledge of tomorrow's realization.

The same logic matters for policy. An announced future tax change may alter consumption today. A change in a policy rule can change expectations and the aggregate relationships observed under the old rule. This is the concern in the Lucas critique.[1] Writing down individual decisions makes these responses explicit. It does not, by itself, guarantee that the chosen preferences, constraints, or information assumptions remain appropriate under every policy change.

1.3 Reading a dynamic equilibrium

The household takes income and returns as given. In general equilibrium, they must be consistent with production and other agents' choices. A competitive equilibrium combines individual optimality, budget constraints, market clearing, and policy rules at each date and, under uncertainty, each history.

States, choices, and paths. Consider a deterministic economy with fixed labor normalized to one, output $f(k_t)$, and no government or foreign trade. Capital k_t is available at the start of period t. With depreciation δ, feasibility gives

$$k_{t+1} = f(k_t) + (1 - \delta)k_t - c_t. \tag{9}$$

Capital is inherited; consumption is chosen and affects tomorrow's capital. Here k_t is a state variable and c_t a control variable. The set of states depends on the model's timing and assumptions.

If households can save in capital, competitive factor pricing gives the gross return $f'(k_{t+1}) + 1 - \delta$. Substituting this return into (4) yields the intertemporal condition

$$u'(c_t) = \beta\,[f'(k_{t+1}) + 1 - \delta]\,u'(c_{t+1}). \tag{10}$$

Together, (9) and (10) describe feasible accumulation and optimal consumption growth. Given k_0, different initial choices of c_0 generate different candidate paths. Feasibility and the Euler equation alone do not establish that each path is an equilibrium. The model also needs its boundary conditions, including the

[1] Robert E. Lucas, Jr. (1976), "Econometric Policy Evaluation: A Critique," *Carnegie-Rochester Conference Series on Public Policy*, 1, 19–46. https://doi.org/10.1016/S0167-2231(76)80003-6.

appropriate transversality condition for this infinite-horizon saving problem. We derive these conditions in the Ramsey chapter.

A steady state is a particular path. For a positive stationary allocation, set $k_{t+1} = k_t = k^*$ and $c_{t+1} = c_t = c^*$ in the two equations. Since $u'(c^*) > 0$,

$$c^* = f(k^*) - \delta k^*, \tag{11}$$

$$1 = \beta\left[f'(k^*) + 1 - \delta\right]. \tag{12}$$

These equations locate a steady state. They do not describe the transition from an arbitrary k_0, or prove existence or uniqueness of an equilibrium path. With sustained technological or population growth, aggregate quantities may keep growing on a balanced growth path. A stationary system may then require normalization, for example by effective labor, rather than constant levels.

Equilibrium and welfare answer different questions. A planner chooses a feasible allocation directly; a competitive equilibrium also specifies prices that support individual choices. Under suitable conditions, a planner's allocation can be decentralized. Taxes, market power, externalities, incomplete markets, or nominal rigidities may prevent that equivalence. Market clearing alone establishes neither efficiency nor the desirability of a policy. A welfare comparison must state whose utility is evaluated, which constraints apply, and what alternative is feasible.

1.4 Accounting and notation used in these notes

We need only a few accounting conventions at the outset. In a closed economy, using gross investment I_t,

$$Y_t = C_t + I_t + G_t, \tag{13}$$

$$K_{t+1} = (1 - \delta)K_t + I_t. \tag{14}$$

K_t is the stock available at the beginning of the period; I_t is a flow during the period. Depreciation already enters (14). We do not add it again to (13). A household's purchase of a financial claim need not create physical capital; domestic claims and liabilities must be consolidated when deriving aggregate feasibility.

Accounting becomes a model only after we specify behavior, technology, and equilibrium. Government budgets enter with fiscal policy; international accounts enter in Part IV.

Capital letters generally denote aggregates; lower-case letters denote per-person or normalized quantities. Check each chapter's normalization before comparing equations. Unless stated otherwise, β is a discount factor, ρ a rate of time preference, r a net real interest rate, and $R = 1 + r$ a gross real return. In discrete time, the convention $\beta = 1/(1 + \rho)$ differs from continuous-time discounting $e^{-\rho t}$.

For a nominal one-period return $1 + i_t$, realized inflation $\pi_{t+1} = P_{t+1}/P_t - 1$ gives the gross real return

$$R_{t+1} = \frac{1 + i_t}{1 + \pi_{t+1}}. \tag{15}$$

An interest rate quoted in nominal units is therefore not the real return in (8). Under uncertainty, the realized price change must remain inside the relevant expectation.

Finally, for a positive variable X_t and a positive stationary reference $\bar{X}$, define

$$\hat{x}_t \equiv \log X_t - \log \bar{X} \approx \frac{X_t - \bar{X}}{\bar{X}}. \tag{16}$$

The definition is exact; the approximation is local. A deviation from a steady state, a growth rate, and an output gap relative to a flexible-price allocation are different objects. Each chapter states its reference point before using an approximation.

Part II
Growth

How does an economy's productive capacity evolve? We begin with growth facts, then move from an imposed saving rate to optimal saving, overlapping generations, and endogenous technological change.

2 Growth Facts and Growth Accounting

Why are some economies richer than others, and why do living standards grow? Before choosing a model, we need to separate the outcomes to be explained. This chapter distinguishes income levels from growth rates, then decomposes output growth into measured inputs and productivity. The decomposition helps locate a question; explaining the underlying decisions requires a growth model.

2.1 Income levels, growth rates, and convergence

2.1.1 Which output, and per whom?

Let Y be real GDP, N the population, E the number of employed people, and L total hours worked. For positive quantities,

$$\frac{Y}{N} = \frac{Y}{L}\frac{L}{E}\frac{E}{N}. \tag{17}$$

Output per person therefore depends on output per hour, hours per employed person, and the employment-to-population ratio. Higher GDP growth need not mean faster growth in GDP per person. Likewise, a rise in GDP per person can come from more work per person without any rise in output per hour.

Use real output to study changes over time. For cross-country level comparisons, purchasing power parities (PPPs) adjust for differences in price levels; conversion at market exchange rates can mix output differences with price and exchange-rate movements.[2] Use a common measure and specify the comparison year. GDP per person measures average production, not its distribution or welfare: leisure, health, and consumption opportunities also matter.

2.1.2 A level difference is not a growth-rate difference

For a positive continuous-time variable $X(t)$, define its instantaneous growth rate as

$$g_X(t) \equiv \frac{\dot{X}(t)}{X(t)} = \frac{d\log X(t)}{dt}. \tag{18}$$

Logarithms turn products into sums. Applying this definition to (17) gives the exact identity

$$g_{Y/N} = g_{Y/L} + g_{L/E} + g_{E/N}. \tag{19}$$

The ratios must be defined consistently; employment growth and population growth do not enter in the same way.

For observations at dates t and $t+T$, the average annual log growth rate is

$$\bar{g}_X = \frac{\log X_{t+T} - \log X_t}{T}, \qquad \frac{X_{t+T}}{X_t} = e^{T\bar{g}_X}. \tag{20}$$

The corresponding compound annual percentage change is $e^{\bar{g}_X} - 1$, approximately $\bar{g}_X$ for small rates. Unless stated otherwise, finite-period accounting below uses log changes, $\Delta \log X_t = \log X_t - \log X_{t-1}$.

[2] World Bank, "ICP 2021: Concepts and Definitions," explains PPP-based comparisons of expenditure volumes. https://www.worldbank.org/en/brief/2021/04/30/icp2021-concepts-definitions.

At a constant growth rate $g > 0$, doubling takes $T = \log 2/g$. A 2% annual log growth rate doubles output in about 35 years. A one-off 10% increase in output raises its level; it leaves subsequent growth unchanged if the economy then returns to its previous growth rate. A lasting increase in the growth rate keeps widening the gap relative to the previous path.

Facts that guide the models. Jones (2016) documents three useful patterns.[3]

- Sustained growth is a feature of modern economic history. US GDP per person grew at about 2% per year over 1870–2007, with substantial variation across shorter periods.
- Income levels differ widely across countries. Some economies catch up; others remain far behind or lose ground over particular periods.
- A roughly stable capital-output ratio is a useful long-run benchmark. Factor shares and growth rates can change, so balanced growth is an approximation to be assessed, rather than a universal empirical law.

A theory must distinguish growth at the frontier from movement toward it. Matching a long-run average also does not require every shock to be temporary.

2.1.3 What does convergence mean?

Consider a poorer economy i and a richer economy j, with per-person outputs y_i and y_j measured in comparable units. Their log income gap evolves as

$$D(t) \equiv \log y_j(t) - \log y_i(t), \tag{21}$$

$$\dot{D}(t) = g_{y_j}(t) - g_{y_i}(t), \tag{22}$$

$$D(T) - D(0) = \int_0^T \left[g_{y_j}(t) - g_{y_i}(t) \right] dt. \tag{23}$$

The relative gap narrows when the poorer economy grows faster over the interval. Positive growth in both economies is insufficient. Nor does a narrowing relative gap necessarily reduce the absolute difference $y_j - y_i$.

Unconditional convergence means poorer economies tend to grow faster without conditioning on other characteristics. *Conditional convergence* allows economies to approach different long-run paths: an economy further below its own path tends to grow faster, holding the determinants of that path fixed. Saving, population growth, and productivity may differ across economies. Diminishing returns to capital alone therefore do not imply that every poor country must catch up with every rich country.[4]

A tendency for initially poorer economies to grow faster is called β-convergence. A decline in the cross-country dispersion of log income is called σ-convergence. These are different statements: new shocks can increase dispersion even when growth tends to be faster at lower initial income. The sample and observation period are part of any convergence claim.

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[3] Charles I. Jones (2016), "The Facts of Economic Growth," *Handbook of Macroeconomics*, 2A, 3–69, especially Sections 1, 2, and 4. The empirical statements here describe that historical evidence, not a current data release. https://doi.org/10.1016/bs.hesmac.2016.03.002.

[4] Robert J. Barro, N. Gregory Mankiw, and Xavier Sala-i-Martin (1992), "Capital Mobility in Neoclassical Models of Growth," NBER Working Paper 4206, discusses convergence relative to each economy's steady state. https://doi.org/10.3386/w4206.

2.2 Capital, labor, and productivity

2.2.1 From a production function to growth accounting

Start with a one-good aggregate production function in continuous time:

$$Y(t) = B(t)F\big(K(t), H(t)\big), \qquad H(t) = h(t)L(t). \tag{24}$$

K is productive physical capital, L hours worked, and h average labor quality per hour. Thus H is quality-adjusted labor input. B is Hicks-neutral total factor productivity (TFP). It multiplies the whole production function; its growth need not equal the growth of labor-augmenting technology used in the Solow chapter.

Assume positive inputs, differentiable F, constant returns to scale, and no additional inputs. The total derivative is

$$\dot{Y} = \dot{B}F + BF_K\dot{K} + BF_H\dot{H}. \tag{25}$$

Divide by $Y = BF$ and multiply and divide each input term by its input level:

$$\frac{\dot{Y}}{Y} = \frac{\dot{B}}{B} + \frac{BF_K K}{Y}\frac{\dot{K}}{K} + \frac{BF_H H}{Y}\frac{\dot{H}}{H}, \tag{26}$$

$$g_Y = g_B + \varepsilon_K g_K + \varepsilon_H g_H, \quad \varepsilon_K \equiv \frac{BF_K K}{Y}, \quad \varepsilon_H \equiv \frac{BF_H H}{Y}. \tag{27}$$

The weights are output elasticities. Constant returns implies $F_K K + F_H H = F$, so $\varepsilon_K + \varepsilon_H = 1$.

To replace elasticities by observed income shares, we need further assumptions. With competitive factor markets and no wedges, let q be the real rental price of capital services and w_H the price of one unit of quality-adjusted labor. Profit maximization gives

$$q = BF_K, \qquad w_H = BF_H. \tag{28}$$

Consequently,

$$s_K \equiv \frac{qK}{Y} = \varepsilon_K, \qquad s_H \equiv \frac{w_H H}{Y} = \varepsilon_H = 1 - s_K. \tag{29}$$

Here q is a gross rental price before depreciation, not the net interest rate. Labor compensation is $w_H H$; with common quality h, the wage per hour is $w_H h$. In this benchmark all output is paid to these two inputs. Pure profits, markups, taxes, or omitted factors can break the equality between measured income shares and output elasticities.

Substitute the shares and use $g_H = g_h + g_L$:

$$\boxed{g_Y = g_B + s_K g_K + (1 - s_K)(g_h + g_L).} \tag{30}$$

This is an exact instantaneous decomposition under the stated production and pricing assumptions. It does not yet determine any input's growth rate.

2.2.2 Labor productivity and finite-period contributions

Subtract g_L from both sides of (30):

$$g_{Y/L} = g_B + s_K g_K + (1 - s_K)g_h - s_K g_L, \tag{31}$$

$$= g_B + s_K g_{K/L} + (1 - s_K)g_h. \tag{32}$$

Output per hour can rise through capital deepening, better labor quality, or higher TFP. Labor productivity Y/L and TFP B are therefore different objects. Combining this result with (19) also separates productivity growth from changes in work per person.

For annual data, a common implementation uses average adjacent-year income shares, $\bar{s}_{K,t} = (s_{K,t} + s_{K,t-1})/2$:

$$\Delta \log Y_t \approx \bar{s}_{K,t} \Delta \log K_t + (1 - \bar{s}_{K,t}) \Delta \log H_t + \Delta \log B_t, \tag{33}$$

$$\Delta \log B_t^{\text{res}} \equiv \Delta \log Y_t - \bar{s}_{K,t} \Delta \log K_t - (1 - \bar{s}_{K,t}) \Delta \log H_t. \tag{34}$$

The first line approximates the continuous-time decomposition over an interval. The second defines the measured residual exactly for the chosen series and weights. It need not equal a pure technology change. For Cobb–Douglas production with constant exponents, the corresponding log decomposition is exact over a finite interval.

An accounting example. Suppose annual log changes are 0.04 for output, 0.06 for capital, and 0.01 for quality-adjusted labor, with capital share $1/3$. Then

$$\Delta \log B^{\text{res}} = 0.04 - \frac{1}{3}(0.06) - \frac{2}{3}(0.01) = 0.013\overline{3}.$$

Capital contributes 2 percentage points, labor about 0.67, and the residual about 1.33, when log changes are multiplied by 100. These sum to 4. This is an illustration, not an estimate for a country. The calculation does not imply that raising investment would cause exactly 2 percentage points of growth.

2.2.3 What is inside the residual?

The measured residual depends on what the inputs capture. Capital services can change differently from the value of the capital stock. Hours need not track employment, and labor quality need not be constant. Unmeasured utilization, input quality, reallocation, and measurement error can all enter the residual.[5] For example, if true labor input is hL but the calculation uses only L, then (30) implies

$$g_B^{\text{res}} = g_Y - s_K g_K - (1 - s_K) g_L = g_B + (1 - s_K) g_h. \tag{35}$$

Omitted improvements in labor quality appear as TFP growth in that calculation. A change in the residual therefore calls for a mechanism and measurement checks before it is interpreted as innovation or technological regress.

2.3 From accounting to a growth model

2.3.1 Level accounting and joint determination

For a transparent benchmark, assume

$$Y = BK^{\alpha}(hL)^{1-\alpha}, \qquad 0 < \alpha < 1. \tag{36}$$

[5] OECD (2001), *Measuring Productivity: OECD Manual*, especially Chapters 2, 4, and 5, on interpretation, labor input, and capital services. https://doi.org/10.1787/9789264194519-en.

Divide by L and define $y \equiv Y/L$ and $k \equiv K/L$:

$$y = Bk^{\alpha}h^{1-\alpha}. \tag{37}$$

For two economies using the same α and comparable measurements,

$$\frac{y_i}{y_j} = \frac{B_i}{B_j}\left(\frac{k_i}{k_j}\right)^{\alpha}\left(\frac{h_i}{h_j}\right)^{1-\alpha}, \tag{38}$$

$$\log\frac{y_i}{y_j} = \log\frac{B_i}{B_j} + \alpha\log\frac{k_i}{k_j} + (1-\alpha)\log\frac{h_i}{h_j}. \tag{39}$$

This decomposes a level gap; it is distinct from accounting for growth over time. B_i/B_j is inferred after choosing the production function and measuring the inputs. The decomposition does not establish three independent causes.

Higher productivity can raise the return to investment and induce more capital accumulation. Its effects may then appear in both the productivity and capital terms. Similarly, education, saving, and technology adoption may respond to common policies or institutions. An accounting contribution records where output differences appear; a causal explanation specifies what changes first and how the other variables respond.

2.3.2 Connecting TFP to labor-augmenting technology

The next chapter writes technology as $Y = F(K, AL)$. To connect the two conventions, set $h = 1$ in (36) and use Cobb–Douglas production:

$$Y = BK^{\alpha}L^{1-\alpha}, \tag{40}$$

$$Y = K^{\alpha}(AL)^{1-\alpha} = A^{1-\alpha}K^{\alpha}L^{1-\alpha}. \tag{41}$$

The two representations describe the same technology when

$$B = A^{1-\alpha}, \qquad g_B = (1-\alpha)g_A. \tag{42}$$

This equivalence uses the Cobb–Douglas form and constant α; it is not a general conversion for every production function.

Suppose a balanced growth path has constant K/Y, constant $h = 1$, and labor growth n. A constant capital-output ratio implies $g_K = g_Y$. Substitution into (30), where $s_K = \alpha$, gives

$$g_Y = g_B + \alpha g_Y + (1-\alpha)n, \tag{43}$$

$$(1-\alpha)(g_Y - n) = g_B, \tag{44}$$

$$g_{Y/L} = \frac{g_B}{1-\alpha} = g_A. \tag{45}$$

Capital deepening accompanies growth on this path. It does not follow that capital accumulation supplies an independent permanent growth engine. We have only derived what the growth rates must satisfy *if* this path exists.

What a growth model must add. Growth accounting leaves saving, capital accumulation, labor supply, and productivity unexplained. Solow begins with a given saving rate and technology path. Ramsey makes saving an intertemporal choice; OLG changes who saves and when; endogenous growth models study the forces sustaining productivity or other reproducible inputs.

For Solow's closed one-good economy without government, feasibility and a constant saving rate s imply

$$Y = C + I, \qquad I = sY, \qquad \dot{K} = sY - \delta K. \tag{46}$$

Here I is gross investment and δ is the continuous-time depreciation rate. Together with production, population growth, and a technology path, these assumptions determine capital dynamics from an initial stock. The next chapter uses that system to ask when the economy converges, what changes its long-run income level, and what can sustain growth in output per person.

3 Solow Model

Growth accounting separates capital accumulation from productivity growth. Solow adds a mechanism: some output becomes new capital, which changes future production. We first ask why this mechanism was worth isolating, then derive its dynamics and examine what it can explain.

3.1 From the growth question to the model

3.1.1 Start with a puzzle, before choosing a model

Saving uses output today and finances capital that can produce tomorrow. Meanwhile, the labor force grows at a rate that need not depend on saving. It seems possible to keep accumulating machines without providing the right amount of equipment for the workers who arrive. What, if anything, brings these two processes into line?

Solow's starting point was a theoretical tension in the Harrod–Domar framework. In his later account, he singled out treating the saving rate, labor-force growth, and the capital-output ratio as independent constants. His 1956 paper asked whether the resulting knife-edge depended on fixed production proportions.[6] The sequence below works through that tension as a modeling exercise. It is a reconstruction from the published argument, not a record of his private thoughts or the order in which he discovered the model.

3.1.2 Try a simple account, and see where it becomes restrictive

Start without technical change. A single good lets us measure output and investment in common units; let K be the stock built from past investment. Give saving a fixed share s, rather than trying to explain every household's choice immediately. With gross output and explicit depreciation, accumulation is $\dot{K} = sY - \delta K$.

This tracks resources, but it does not tell us how much output the stock can produce. A simple next attempt is a fixed production recipe:

$$Y = \min\left\{\frac{K}{v}, \frac{L}{a}\right\}, \qquad v, a > 0. \tag{47}$$

A unit of output requires v units of capital and a units of labor. If both inputs are fully used, $K = vY$ and $L = aY$. Substituting the first relation into accumulation gives

$$\frac{\dot{K}}{K} = s\frac{Y}{K} - \delta = \frac{s}{v} - \delta. \tag{48}$$

Full use of both inputs also requires capital and output to grow at the labor-force growth rate n. Hence

$$\frac{s}{v} - \delta = n. \tag{49}$$

Now the tension is visible. Nothing in the saving rule makes its left-hand side equal the growth of labor on the right. A small parameter change can make this fully utilized growth path impossible under the fixed recipe. This example isolates the compatibility problem; it does not reproduce every part of Harrod's or Domar's analysis.

[6] Robert M. Solow (1956), "A Contribution to the Theory of Economic Growth," *Quarterly Journal of Economics*, 70(1), 65–94, Sections I–III, https://doi.org/10.2307/1884513; and Solow (1987), "Growth Theory and After," Nobel lecture, https://www.nobelprize.org/prizes/economic-sciences/1987/solow/lecture/.

3.1.3 Ask which restriction is doing the work

Before concluding that growth is intrinsically unstable, examine what made the equality necessary. We required every unit of output to use the same amount of capital and labor. Extra machines could not economize on workers, and extra workers could not use a less capital-intensive technique.

There are several possible responses. Saving might adjust; labor supply or employment might change; firms might use different factor proportions. These responses would produce different models. Solow kept a simple saving rule and an exogenous labor supply, allowed substitution in production, and studied paths with full employment. This makes the production restriction the central experiment. Full employment is maintained in that experiment, rather than proved to hold in the actual economy.

Replace the fixed recipe by $Y = F(K, L)$. More machines per worker can now raise output, while diminishing returns reduce output per unit of capital. Thus Y/K can fall as capital accumulates. In $\dot{K}/K = sY/K - \delta$, capital growth can slow toward labor-force growth without a change in s or n. This is a candidate adjustment mechanism. Whether it works still depends on the properties of F; the dynamics below establish the result.

3.1.4 Turn the question into variables and a closed system

Choosing a flexible F is not enough. In $\dot{K} = sF(K, L) - \delta K$, both K and L could vary. To study capital adjustment first, specify $L(t) = L_0 e^{nt}$ and give $K(0)$. This leaves one inherited stock to solve for. An alternative with wage-sensitive labor supply would require another decision rule; it is a possible extension, not an algebraic rearrangement.

Next, a large economy has more capital partly because it has more workers. The relevant object for this question is K/L. Constant returns allows $Y/L = F(K/L, 1)$, reducing the accumulation problem to a single capital intensity. The ratio is selected because it removes scale while preserving the mechanism we want to study.

Finally, if labor efficiency rises, even constant equipment per worker may fail to keep up with productive opportunities. Introduce labor-augmenting technology A and use $K/(AL)$ instead. We can then distinguish accumulating capital relative to effective labor from growth in output per worker. We take A as given at this stage; explaining innovation would add a different question.

The model has now acquired its structure through successive choices: an accounting constraint, a production experiment, a saving rule, an exogenous labor path, and a useful normalization. The assumptions that follow collect these choices so that we can check which conclusion depends on which one.

3.2 Economic environment and assumptions

The map below separates what the model takes as given, how agents behave, and which conditions make their decisions consistent. The paragraphs that follow explain the assumptions and fix the notation.

01 Economy

A closed, one-good economy. No government or foreign trade; output can be consumed or invested.

02 Technology

Constant returns in capital and effective labor; diminishing returns to capital intensity.

$$Y_t = F(K_t, A_t L_t).$$

03 Households

Own capital and supply labor inelastically. Saving is a given rule, rather than an optimal choice:

$$S_t = sY_t, \qquad 0 < s < 1.$$

04 Firms

Choose capital and labor to maximize profits at competitive factor prices:

$$\max_{K,L} \; F(K, A_t L) - q_t K - w_t L.$$

05 Given states and paths

Positive initial stocks. Capital is inherited; technology and labor follow exogenous paths:

$$\frac{A_{t+1}}{A_t} = 1 + g, \quad \frac{L_{t+1}}{L_t} = 1 + n.$$

06 Consistency and timing

Goods and factor markets clear. Investment becomes usable next period:

$$Y_t = C_t + I_t, \qquad I_t = S_t,$$
$$K_{t+1} = (1 - \delta)K_t + I_t.$$

$$\underbrace{K_t, A_t, L_t}_{\text{Given at } t} \longrightarrow \underbrace{Y_t \;\to\; C_t, I_t}_{\text{Production and allocation}} \longrightarrow \underbrace{K_{t+1}}_{\text{Available at } t+1} .$$

3.2.1 Resources, timing, and behavior

The economy is closed, has no government, and produces one good. Households own capital and supply labor inelastically. Each worker supplies one unit of labor; changes in hours, participation, and employment rates are omitted. Firms hire labor and rent capital in competitive markets.

In discrete time, K_t is available at the start of period t. Given K_t, A_t, and L_t, firms produce Y_t. Households consume part of output and save the fraction $s \in (0, 1)$. Investment becomes available next period:

$$Y_t = C_t + I_t, \qquad I_t = S_t = sY_t, \tag{50}$$
$$K_{t+1} = (1 - \delta)K_t + sY_t, \qquad 0 \le \delta \le 1. \tag{51}$$

The saving rule is imposed. Firms still maximize profits; the absence of a household optimization problem does not imply that firms are non-optimizing.

Technology and labor follow

$$A_{t+1} = (1 + g)A_t, \qquad L_{t+1} = (1 + n)L_t, \qquad g, n \ge 0. \tag{52}$$

Initial K_0, A_0, L_0 are positive. We assume $\delta + n + g > 0$ to obtain a finite positive steady state below. Technology is labor-augmenting: an increase in A makes a worker provide more effective labor. Its availability and growth are given, rather than explained by research or adoption decisions.

3.2.2 Production and normalization

Let

$$Y_t = F(K_t, A_t L_t). \tag{53}$$

F has constant returns in capital and effective labor:

$$F(\lambda K, \lambda AL) = \lambda F(K, AL), \qquad \lambda > 0. \tag{54}$$

Replicating all inputs replicates output. This rules out a fixed factor, such as land, that becomes scarcer as the economy expands. It also rules out aggregate increasing returns in these two inputs.

Divide by effective labor and define

$$k_t \equiv \frac{K_t}{A_t L_t}, \qquad y_t \equiv \frac{Y_t}{A_t L_t} = F(k_t, 1) \equiv f(k_t), \qquad c_t \equiv \frac{C_t}{A_t L_t} = (1 - s) f(k_t). \tag{55}$$

Here lower-case variables are *per effective worker*. Output per worker is $Y_t/L_t = A_t y_t$, and capital per worker is $K_t/L_t = A_t k_t$. This differs from the per-hour normalization used for level accounting in Chapter 2. With $A_t = 1$, the two capital normalizations coincide.

Assume $f(0) = 0$ and, for $k > 0$,

$$f'(k) > 0, \qquad f''(k) < 0, \qquad \lim_{k \downarrow 0} f'(k) = \infty, \qquad \lim_{k \to \infty} f'(k) = 0. \tag{56}$$

The last two are the Inada conditions on *marginal products*, not output levels. With the other assumptions, they make capital sufficiently productive near zero and insufficiently productive at very high intensity to guarantee an interior crossing of investment and replacement needs. They are sufficient benchmark restrictions, not empirical laws for every technology.

Our running example is Cobb–Douglas:

$$F(K, AL) = K^{\alpha} (AL)^{1-\alpha}, \qquad f(k) = k^{\alpha}, \qquad 0 < \alpha < 1. \tag{57}$$

Diminishing returns here means adding machines to a fixed amount of effective labor produces progressively smaller additions to output. It does not mean that output falls when capital rises.

3.2.3 Prices and market clearing

At each date, let q_t be the real rental price of capital and w_t the wage per worker. A price-taking firm solves

$$\max_{K, L} \; F(K, A_t L) - q_t K - w_t L. \tag{58}$$

At an interior solution, denoting derivatives with respect to the two arguments of F by F_1, F_2,

$$q_t = F_1(K_t, A_t L_t), \qquad w_t = A_t F_2(K_t, A_t L_t). \tag{59}$$

Constant returns and Euler's theorem for homogeneous functions give

$$Y_t = F_1 K_t + F_2 A_t L_t = q_t K_t + w_t L_t. \tag{60}$$

Economic profits are zero after paying both factors, including capital owned by households. This is a property of this competitive constant-returns benchmark, not a claim that owners receive no capital income.

For Cobb–Douglas, $q_t = \alpha k_t^{\alpha-1}$ and $w_t = (1-\alpha)A_t k_t^{\alpha}$. Capital deepening lowers the rental return and raises the wage per effective worker. Prices support full use of the supplied capital and labor. Goods-market clearing and the saving rule then determine investment. Given the initial stocks and exogenous paths, the model is closed.

The mechanism in one loop. Capital and effective labor determine output; the saving rule turns some output into investment; investment and depreciation determine next period's capital. Factor prices describe the returns along this path. The saving rate does not respond to those returns until we change the household side of the model.

3.3 Solow model in discrete time

3.3.1 Deriving the law of motion

Define $D \equiv (1+n)(1+g)$, the gross growth factor of effective labor. Divide (51) by $A_{t+1}L_{t+1}$:

$$k_{t+1} = \frac{(1-\delta)K_t + sY_t}{A_{t+1}L_{t+1}}, \tag{61}$$

$$= \frac{(1-\delta)K_t/(A_tL_t) + sY_t/(A_tL_t)}{D}, \tag{62}$$

$$= \frac{(1-\delta)k_t + sf(k_t)}{D} \equiv \Phi(k_t). \tag{63}$$

The numerator adds surviving capital and investment per current effective worker. The denominator spreads it across next period's effective labor. New workers need equipment, and increasing labor efficiency requires more equipment per worker to keep k constant.

Subtract k_t and define $d_D \equiv D - 1 + \delta$:

$$k_{t+1} - k_t = \frac{sf(k_t) - d_D k_t}{D}, \qquad d_D = \delta + n + g + ng. \tag{64}$$

Break-even investment $d_D k_t$ keeps capital per effective worker unchanged. The ng term is part of the exact discrete-time expression; dropping it is a small-rate approximation. Setting $g = 0$ gives $d_D = \delta + n$; setting $g = n = 0$ gives $d_D = \delta$.

Reading the per-worker version. Let $\bar{k}_t = K_t/L_t$. Dividing the aggregate accumulation equation by L_{t+1} instead gives the exact law

$$\bar{k}_{t+1} = \frac{(1-\delta)\bar{k}_t + sF(\bar{k}_t, A_t)}{1+n}.$$

There is no $1+g$ in this denominator: technology remains inside production. With constant $A_t = \bar{A}$, define $\bar{f}(\bar{k}) = F(\bar{k}, \bar{A})$. The map is then stationary; setting $n = 0$ also removes the population-growth denominator. These are valid restricted cases. When technology grows, a stationary map in $k = K/(AL)$ requires the effective-labor normalization above.

3.3.2 A steady state and the path toward it

A positive normalized steady state satisfies

$$k^* = \Phi(k^*), \qquad sf(k^*) = d_D k^*, \qquad \frac{K}{Y} = \frac{k^*}{f(k^*)} = \frac{s}{d_D}. \tag{65}$$

A stable capital-output ratio is a conclusion *on this path*, not something that follows from labor-augmenting notation alone.

For existence and uniqueness, divide the middle equation by $k^* > 0$. Strict concavity and $f(0) = 0$ imply $f'(k) < f(k)/k$, so

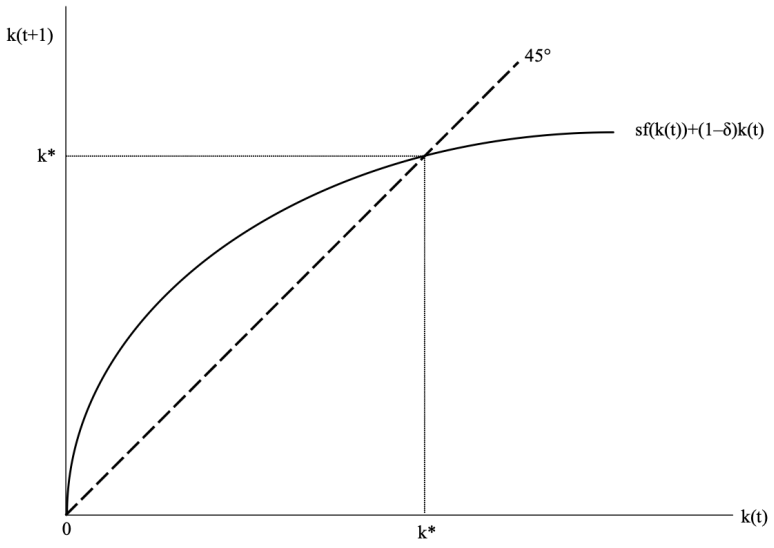
$$\frac{d}{dk}\frac{f(k)}{k} = \frac{kf'(k) - f(k)}{k^2} < 0. \tag{66}$$

The Inada conditions imply $f(k)/k$ falls from infinity to zero. Hence $sf(k)/k = d_D > 0$ has exactly one positive solution. Zero is also an absorbing boundary under $f(0) = 0$; our convergence result starts from $k_0 > 0$.

Below k^*, saving exceeds break-even investment and k rises. Above k^*, saving is insufficient to maintain that intensity and k falls. Moreover,

$$\Phi'(k) = \frac{1 - \delta + sf'(k)}{D} > 0. \tag{67}$$

Since Φ is increasing and $\Phi(k^*) = k^*$, an initial $k < k^*$ maps below k^*, while an initial $k > k^*$ maps above it. The sequence is therefore monotone and bounded toward k^*, and continuity makes its limit a fixed point. The unique positive fixed point is the limit for every $k_0 > 0$.



Reading the diagram. This diagram shows the special case $n = g = 0$, with $\delta > 0$. The curved map is next period's capital; the 45-degree line is unchanged capital. Starting on the horizontal axis, move vertically to the map, then horizontally to the 45-degree line to carry that value into the next iteration. For $n > 0$ or $g > 0$, divide the map by D as in (63).

A steady state is one particular equilibrium path. A transition from an arbitrary initial stock is also an equilibrium path under the imposed saving rule. Convergence does not make the transition cease to be an equilibrium.

3.3.3 What grows on a balanced growth path?

If $k_t = k^*$, then $y_t = f(k^*)$ and $c_t = (1-s)f(k^*)$ are constant. Yet

$$\frac{Y_t}{L_t} = A_t f(k^*), \qquad \frac{K_t}{L_t} = A_t k^*, \qquad Y_t = A_t L_t f(k^*). \tag{68}$$

Output and capital per worker grow at gross rate $1+g$; their aggregates grow at D. With no technical progress, there is no sustained growth per worker on this path, even if the population and total output keep growing.

For Cobb–Douglas, solve $s(k^*)^\alpha = d_D k^*$:

$$k^* = \left(\frac{s}{d_D}\right)^{1/(1-\alpha)}, \qquad y^* = \left(\frac{s}{d_D}\right)^{\alpha/(1-\alpha)}. \tag{69}$$

A higher saving rate raises the long-run level of output per effective worker. Holding the technology path fixed, it does not change the long-run growth rate of output per worker. Its steady-state level elasticity is

$$\frac{\partial \log y^*}{\partial \log s} = \frac{\alpha}{1-\alpha}. \tag{70}$$

This comparison holds other determinants fixed. It is not a causal estimate from a cross-country correlation between saving and income.

3.3.4 Saving, rebuilding, and conditional convergence

Suppose an economy starts on its old balanced growth path and permanently raises s. At the time of the change, K, A, and L are inherited, so output is unchanged and consumption falls. Higher investment raises future K; output per worker then grows faster during the transition. As diminishing returns reduce the extra investment generated by another unit of capital, k approaches its new, higher steady state. Long-run growth per worker returns to g.

This explains why an investment expansion can have a lasting level effect without a lasting growth-rate effect. It does not establish that any observed investment campaign will be productive: the one-good benchmark abstracts from project selection, idle capacity, financing, and heterogeneous returns.

A one-time loss of physical capital, with technology, labor, and saving behavior unchanged, lowers k. The high marginal product at the lower stock supports rebuilding toward the same path. Actual disasters can also destroy skills, institutions, and technology, so this experiment isolates a capital-loss mechanism rather than predicting every recovery.

The convergence result is conditional on the determinants of k^* and the technology path. A low-income economy with a low saving rate or low productivity need not grow faster than a richer economy already close to a higher path. Income alone does not tell us distance from the relevant steady state.

3.3.5 Local convergence speed

Let $x_t = k_t - k^*$. A first-order Taylor expansion at k^* gives

$$k_{t+1} = \Phi(k^*) + \Phi'(k^*)x_t + O(x_t^2), \tag{71}$$

$$x_{t+1} \approx \phi x_t, \qquad \phi \equiv \frac{1 - \delta + sf'(k^*)}{D}. \tag{72}$$

Because $sf'(k^*) < sf(k^*)/k^* = d_D$, we have $0 < \phi < 1$. Iterating the linearized system yields

$$x_t \approx \phi^t x_0, \qquad \lambda_D = -\log\phi > 0, \qquad T_{1/2} \approx \frac{\log 2}{\lambda_D}. \tag{73}$$

ϕ is persistence; a larger ϕ means slower adjustment. λ_D is the corresponding decay rate per period, and $T_{1/2}$ is the local half-life of a deviation. The recursion for the linearized system has an exact geometric solution; the approximation is in using that system to describe the original model. The nonlinear path need not have this half-life far from its steady state.

For Cobb–Douglas, $sf'(k^*) = \alpha d_D$, so

$$\phi = 1 - \frac{(1-\alpha)d_D}{D}. \tag{74}$$

Adjustment is gradual because productive capacity is a stock. Investment can change immediately, but only accumulated investment changes that stock.

3.4 Solow model in continuous time

3.4.1 The same mechanism with continuous adjustment

Now use continuous rates $\dot{L}/L = n$, $\dot{A}/A = g$, and depreciation rate $\delta \geq 0$, with $b \equiv n + g + \delta > 0$. These are rates per unit of time; they are not numerically identical to finite-period percentage changes. Write

$$\dot{K} = sF(K, AL) - \delta K, \qquad k = \frac{K}{AL}, \qquad y = f(k). \tag{75}$$

Log differentiation of the definition of k, followed by substitution, gives

$$\frac{\dot{k}}{k} = \frac{\dot{K}}{K} - \frac{\dot{A}}{A} - \frac{\dot{L}}{L}, \tag{76}$$

$$= \frac{sY}{K} - \delta - g - n, \tag{77}$$

$$\dot{k} = s\frac{Y}{AL} - (n + g + \delta)k = sf(k) - bk \equiv G(k). \tag{78}$$

With $g = 0$ and $A = 1$, this becomes $\dot{k} = sf(k) - (n + \delta)k$. That familiar constant-technology equation is an exact special case.

New investment increases capital intensity; depreciation, population growth, and technical progress reduce it for a given aggregate stock. Technical progress still raises productive possibilities: it reduces k here because each worker now represents more effective labor.

For a short time interval Δt, effective labor grows by approximately $1 + (n + g)\Delta t$ and depreciation is approximately $\delta \Delta t$. Applying the discrete accounting over that interval gives

$$k(t + \Delta t) - k(t) = [sf(k(t)) - bk(t)]\,\Delta t + o(\Delta t). \tag{79}$$

Divide by Δt and take its limit to obtain (78). This explains why the continuous break-even rate is b, whereas the exact discrete expression contains the product ng.

The positive steady state satisfies $sf(k^*) = bk^*$; $G(k) > 0$ below it and $G(k) < 0$ above it. On the balanced growth path, Y/L and K/L grow at g, Y and K grow at $n + g$, and $K/Y = s/b$. Under competitive pricing the net instantaneous real return is $r = f'(k) - \delta$.

3.4.2 Local approximation and an exact Cobb–Douglas solution

Linearizing the law of motion around its steady state gives

$$\dot{k} \approx G'(k^*)(k - k^*), \tag{80}$$

$$G'(k^*) = sf'(k^*) - b < 0, \tag{81}$$

$$k(t) - k^* \approx [k(0) - k^*]e^{-\lambda_C t}, \qquad \lambda_C \equiv b - sf'(k^*) > 0. \tag{82}$$

The exponential expression solves the *linearized* system. It is generally only a local approximation to the original nonlinear equation. For Cobb–Douglas, $\lambda_C = (1 - \alpha)b$ and the local half-life is $\log 2/[(1 - \alpha)b]$.

Cobb–Douglas also permits an exact solution using a different variable. Set $z(t) = k(t)^{1-\alpha}$ and differentiate:

$$\dot{z} = (1 - \alpha)k^{-\alpha}\dot{k}, \tag{83}$$

$$= (1 - \alpha)k^{-\alpha}(sk^{\alpha} - bk) = (1 - \alpha)(s - bz). \tag{84}$$

This is a linear equation in z, with steady state $z^* = s/b$. Its solution and the implied capital path are

$$z(t) = \frac{s}{b} + \left[z(0) - \frac{s}{b}\right]e^{-(1-\alpha)bt}, \tag{85}$$

$$k(t) = \left\{\frac{s}{b} + \left[k(0)^{1-\alpha} - \frac{s}{b}\right]e^{-(1-\alpha)bt}\right\}^{1/(1-\alpha)}. \tag{86}$$

Capital approaches its steady state without arriving there at a finite date unless it starts there. “Time to convergence” must therefore specify a tolerance or a half-life, rather than an exact date of arrival.

3.5 The Golden Rule and the cost of saving

More saving raises steady-state output, but output is not consumption. To maintain a larger capital stock, the economy must devote more resources to investment. The Golden Rule asks which balanced growth path provides the largest consumption per effective worker, holding the technology and labor paths fixed.

In continuous time, a stationary k requires investment bk, so feasible steady-state consumption is

$$c^* = f(k) - bk. \tag{87}$$

Maximize this expression over k:

$$\frac{dc^*}{dk} = f'(k) - b, \tag{88}$$

$$f'(k_{\mathrm{GR}}) = b = n + g + \delta, \qquad \frac{d^2c^*}{dk^2} = f''(k) < 0. \tag{89}$$

One extra unit of capital adds $f'(k)$ to output but requires b units of investment per unit of time to maintain its intensity. At the Golden Rule, these marginal amounts are equal. Equivalently, the net marginal return $f'(k_{\mathrm{GR}}) - \delta$ equals $n + g$.

The saving rate that supports this stock satisfies

$$s_{\mathrm{GR}} = \frac{bk_{\mathrm{GR}}}{f(k_{\mathrm{GR}})}. \tag{90}$$

For Cobb–Douglas, substitute $\alpha k_{\mathrm{GR}}^{\alpha-1} = b$:

$$k_{\mathrm{GR}} = \left(\frac{\alpha}{b}\right)^{1/(1-\alpha)}, \qquad s_{\mathrm{GR}} = \alpha. \tag{91}$$

In discrete time the argument is the same with d_D replacing b: $c^* = f(k) - d_D k$ and $f'(k_{\mathrm{GR}}) = d_D$. The Cobb–Douglas saving share is again α under this timing convention.

> **A welfare boundary.** The Golden Rule maximizes consumption *across balanced growth paths*. With constant technology and strictly increasing utility $u(c)$, maximizing stationary consumption also maximizes utility evaluated at that stationary consumption. Solow has not specified discounting or how to value consumption losses during a transition. It therefore does not establish that moving toward the Golden Rule maximizes intertemporal welfare.

If the initial steady state lies below k_{GR}, raising saving toward s_{GR} increases eventual consumption but requires sacrificing consumption while capital is built. If it lies above k_{GR}, reducing saving toward the Golden Rule can raise both current and eventual consumption. Investment is then maintaining more capital than is useful for maximizing steady-state consumption. A high investment share alone is therefore not a welfare criterion. Ramsey will specify preferences and make the transition trade-off part of the saving decision; OLG will examine dynamic efficiency in an economy with different generations.

3.6 Extensions: which assumption needs to change?

An extension should answer a question left open by the benchmark. Adding an input, changing the investment technology, and explaining productivity are different changes; they need not have the same implications for long-run growth. The examples below use continuous time.

3.6.1 Investment spending and productive capital

In the baseline, one unit of investment creates one unit of productive capital. To separate spending from productive capacity, suppose only a fraction $\eta \in (0, 1]$ becomes usable capital:

$$I = sY, \qquad C = (1 - s)Y, \qquad \dot{K} = \eta I - \delta K. \tag{92}$$

The unconverted part is lost in installation or implementation; it is not additional consumption. With the same labor and technology paths,

$$\dot{k} = \eta s f(k) - bk, \qquad k^* = \left(\frac{\eta s}{b}\right)^{1/(1-\alpha)} \quad \text{for } f(k) = k^\alpha. \tag{93}$$

Higher investment spending and more effective conversion into capital both raise the steady-state stock, but only the former directly reduces the consumption share. Poor project selection or implementation can therefore weaken the link between a reported investment share and productive capacity. This scalar η is an illustrative reduced form: it does not yet explain which projects are chosen or how finance allocates resources.

At a stationary k, maintaining it requires $I/(AL) = bk/\eta$, so $c = f(k) - bk/\eta$. The Golden Rule becomes $f'(k) = b/\eta$. Welfare comparisons must account for the resources spent creating capital, not only the stock that eventually appears in production.

3.6.2 Accumulating human capital

Machines need skills to be useful. The augmented Solow model of Mankiw, Romer, and Weil adds a reproducible human-capital stock to the production function.[7] Write

$$Y = K^\alpha H^\beta (AL)^{1-\alpha-\beta}, \qquad \alpha, \beta > 0, \quad \alpha + \beta < 1. \tag{94}$$

Here H denotes a stock of human capital, rather than the quality-adjusted hours denoted by H in Chapter 2. To keep the distinction visible, use $u \equiv H/(AL)$ for human capital per effective worker. Then

$$y = k^\alpha u^\beta. \tag{95}$$

Suppose output can be transformed into either stock, with fixed investment shares $s_K, s_H > 0$, $s_K + s_H < 1$, and the same depreciation rate for both:

$$\dot{K} = s_K Y - \delta K, \qquad \dot{H} = s_H Y - \delta H, \tag{96}$$

$$\dot{k} = s_K y - bk, \qquad \dot{u} = s_H y - bu. \tag{97}$$

There are now two state variables. Consumption is $c = (1 - s_K - s_H)y$; investment in skills uses resources rather than appearing costlessly in production.

At a positive steady state, the two accumulation equations imply

$$k^* = \frac{s_K}{b} y^*, \qquad u^* = \frac{s_H}{b} y^*. \tag{98}$$

Substitute both into production and collect the powers of y^*:

$$y^* = \left(\frac{s_K y^*}{b}\right)^\alpha \left(\frac{s_H y^*}{b}\right)^\beta, \tag{99}$$

$$(y^*)^{1-\alpha-\beta} = \frac{s_K^\alpha s_H^\beta}{b^{\alpha+\beta}}, \tag{100}$$

$$y^* = \left(\frac{s_K^\alpha s_H^\beta}{b^{\alpha+\beta}}\right)^{1/(1-\alpha-\beta)}. \tag{101}$$

[7] N. Gregory Mankiw, David Romer, and David N. Weil (1992), "A Contribution to the Empirics of Economic Growth," *Quarterly Journal of Economics*, 107(2), 407–437, Section II. https://doi.org/10.2307/2118477.

The condition $\alpha + \beta < 1$ preserves diminishing returns to the two accumulated inputs together. The extension can change income levels and transition dynamics, while long-run growth per worker on its balanced growth path still comes from A. A higher human-capital investment share is not, by itself, a permanent growth-rate increase in this specification.

Years of schooling are a proxy for skills, not a direct observation of H or s_H. Pritchett (2001) highlights why schooling expansion need not translate uniformly into aggregate growth: learning quality, demand for educated labor, and the activities rewarded by institutions can differ.[8] Weak estimates in a particular specification do not establish that skills have no productive value. They call for checking the proposed accumulation mechanism, measurement, and identification.

3.6.3 Technology adoption and catch-up

Giving every economy the same A path assumes away differences in access to and use of technology. To illustrate an adoption mechanism, let frontier efficiency A_F grow at g_F and posit, for a follower with $A \le A_F$,

$$\frac{\dot{A}}{A} = g_F + \mu \log \frac{A_F}{A}, \qquad \mu > 0. \tag{102}$$

This is an illustrative adoption rule, not a consequence of the original saving equation. Define the log technology gap $v = \log(A_F/A)$. Then

$$\dot{v} = g_F - \frac{\dot{A}}{A} = -\mu v, \tag{103}$$

$$v(t) = v(0)e^{-\mu t}. \tag{104}$$

A larger gap creates scope for faster adoption; μ governs how quickly that scope is used. Technology catch-up differs from rebuilding a low capital stock. It can change output directly and also induce capital accumulation.

During catch-up, $\dot{A}/A$ varies with the gap. The capital equation is now

$$\dot{k} = sf(k) - [n + \delta + g_F + \mu v(t)]\, k. \tag{105}$$

We cannot use the earlier constant-g transition unchanged. As the technology gap closes, this equation approaches the benchmark with $g = g_F$. Training, organizational changes, and technology access could help determine μ, but this rule does not model their costs or incentives. Explaining those choices is a further step toward endogenous productivity.

3.6.4 Removing diminishing returns: the AK benchmark

What if returns to the reproducible stock do not diminish? Let

$$Y = \mathscr{A}K, \qquad \mathscr{A} > 0 \text{ constant}, \qquad \dot{K} = (s\mathscr{A} - \delta)K. \tag{106}$$

With labor growth n, output per worker satisfies

$$g_{Y/L} = s\mathscr{A} - \delta - n. \tag{107}$$

[8] Lant Pritchett (2001), "Where Has All the Education Gone?," *World Bank Economic Review*, 15(3), 367–391, especially Section II. https://doi.org/10.1093/wber/15.3.367.

A permanent increase in saving now raises the growth rate permanently. The reason is the changed production assumption: each additional unit of capital has the same product. The Solow Inada and strict-concavity conditions no longer hold, and there is no positive stationary K/L when this growth rate is positive.

K can be interpreted as a broad reproducible composite only if we explain how its components can be accumulated without diminishing aggregate returns. The AK equation shows the role of that restriction; it does not by itself supply a theory of innovation, externalities, or optimal saving. Those mechanisms belong in the endogenous growth chapter.

3.6.5 Fixed land and the transition to modern growth

The baseline omits a fixed input that becomes scarce. Add land X:

$$Y = BK^{a}L^{\ell}X^{1-a-\ell}, \qquad a, \ell > 0, \quad a + \ell < 1. \tag{108}$$

B is Hicks-neutral productivity, as in Chapter 2. Dividing by labor gives

$$\frac{Y}{L} = B\left(\frac{K}{L}\right)^{a}\left(\frac{X}{L}\right)^{1-a-\ell}. \tag{109}$$

At given B and K/L, population growth lowers land per worker and hence output per worker. If population growth also responds to income, productivity gains may support more people rather than persistent gains in living standards. The land equation alone does not establish this outcome; it needs a demographic mechanism.

Hansen and Prescott (2002) study a transition in which a less land-intensive production process becomes profitable as technology advances. Both production processes are available throughout; which one is used changes in equilibrium. Their analysis retains exogenous technical progress, so it explains a transition in production rather than the ultimate source of innovation.[9]

Where the benchmark ends. Lower labor-force growth reduces dilution in Solow, holding everything else fixed. This is not a complete assessment of ageing: participation, pensions, and age-specific saving are absent. Likewise, the model contains no money, banks, or foreign assets. Monetary, financial, or trade extensions need those objects and their decision rules; their effects cannot be inferred merely from their absence here.

The main unresolved choice is already inside the model: why should households save a fixed share when their income and returns change? Ramsey replaces that rule with an intertemporal problem, making consumption, patience, and the return to saving jointly determine the accumulation path.

[9] Gary D. Hansen and Edward C. Prescott (2002), "Malthus to Solow," *American Economic Review*, 92(4), 1205–1217. https://doi.org/10.1257/00028280260344731.

4 Ramsey Model

See Chapter 5 to Chapter 7 from Acemoglu, 2004 for more mathematical help.

We begin the road to sophiscated macro skyscraper with the most basic frame, the Ramsey model.

The Ramsey model[10], or the RCK model, or the Neoclassical growth model differs from the Solow model in that the choice of consumption is explicitly microfounded at a point in time and so endogenizes the savings rate. Ramsey model is in nature a growth model with consumer optmization.

The most intuitive change is now we put S_t under $\max$ as $\max_{\{C_t, S_t\}}$.

4.1 Assumptions

- Same as Solow Model
  - Production Function in the Harrold neutrality form
  - **Exogenous Variables:**
  - Population growth rate= n
  - Technology growth rate= g
  - Depreciation rate= 0 (for simplicity)
  - And most of all, the technological growth rate is assumed as a.
- Firm
  - Perfect competitive firms, or homogenous firms, or representive firms
  - Hire workers and capital in competitive factor markets
  - Each has the production function $Y = F(K, AL)$
  - Firms are owned by the families, meaning the revenue goes straight to latter
- Household
  - Homogenous families.
  - Population growth of each household is assumed as n.
  - Each member in a household provides one unit of labor.
  - Income of a household is the sum of the labor revenue subsum, the capital revenue subsum and the firm revenue share subsum.
  - A household should and must maximize the utility.
  - Degree of patience denoted as discount factor $\beta = \frac{1}{1+\rho}$, and ρ is the discount rate.

Note 1 (Rule of Prohibiting Ponzi Schemes)**.** *Conconstraint in Ramsey model is also called Prohibiting Ponzi Schemes Rule. It forbids the household to do a scam of rolling over debt by repeatedly borrowing. Yet in real life, many people are doing this as a harmless way to ralex debt pressure.*

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[10] Ramsey, Frank P. (1928). "A Mathematical Theory of Saving". Economic Journal. 38 (152): 543–559.

4.2 Solution of Ramsey model

4.2.1 Central Planner's Problem

Originally Ramsey set out the model as a social planner's problem of maximizing levels of consumption over successive generations. Only later was a model adopted by Cass and Koopmans as a description of a decentralized dynamic economy with a representative agent. This is also why the model is called Ramsey-Cass-Koopsman model, yet abbreviated as Ramsey model.

So, chronologically the first thing we should address is central planner's problem.

Disperssive competition

UMP

$$\begin{aligned} &\max_{\{C_t,K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_t) \\ &s.t. C_t + K_{t+1} - (1-\delta)K_t = W_t L_t + R_t k_t + D_t \end{aligned} \tag{110}$$

Noticing that the constraint is equivalent to $C_t + I_t = Y_t + D_t$

Solving this UMP resulting Euler's equation

$$\begin{aligned} &\Rightarrow u'[F(K_t) - K_{t+1}] = u'(C_t) = \beta u'(C_{t+1})F'(K_{t+1}) \\ &\iff \frac{u'(C_t)}{u'(C_{t+1})} = \beta R_{t+1} \end{aligned} \tag{111}$$

PMP at a perfectly competitive market

$$\max_{K_t} \ \pi = F(K_t, A_t L_t) - (r+\delta)K_t - wL_t \tag{112}$$

where the $A_t L_t$ is assumed as a constant in order for model to converge

$$F.O.C. \Rightarrow r + \delta = R = 1 + r \tag{113}$$

In a conclusion, optimization in the disperssive economy yields $\begin{cases} \frac{u'(C_t)}{u'(C_{t+1})} = \beta R_{t+1} \\ r + \delta = R = 1 + r \end{cases}$

Central Planner's Problem

$$\begin{aligned} &\max_{\{C_t,K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_t) \\ &s.t.\ C_t + K_{t+1} = Y_t, \text{ where } \delta \text{ is assumed as 1} \end{aligned} \tag{114}$$

Using Lagranian method and acquire Euler's equation

$$\begin{aligned} &\Rightarrow u'(F(K_t) - K_{t+1}) = u'(C_t) = \beta u'(C_{t+1})F'(K+1) \\ &\iff \frac{u'(C_t)}{u'(C_{t+1})} = \beta R_{t+1} \end{aligned} \tag{115}$$

Conclusion

Regardless of the economy being disperssive or central planning, we would have the same Euler's equation of consumption. This suggests for simplicity, we might as well use central planner's problem.

4.2.2 Method of Lagrangian

We start from scratch, then move on to methods like using Hamiltonian equation or using Bellman equation.

one-period static optimization

$$\begin{aligned} &v = f(x_1, x_2) \\ &s.t.\ g(x_1, x_2) = z \end{aligned} \tag{116}$$

two-period dynamic optimization

A general form is denoted as

$$\begin{aligned} &\max v = f(x_1, x_2) \\ &s.t.\ g(x_1, x_2) = z \text{ (where the subscript is denoted as time periods)} \end{aligned} \tag{117}$$

a two-period consumer maximization problem can be denoted as

$$\begin{aligned} &\max_{C_1, C_2}\ u_1 = u(C_1) + \beta u(C_2) \\ &s.t. \begin{cases} C_1 + S_1 \leqslant [r + (1-\delta)]S_0 + Y_1 \\ C_2 + S_2 \leqslant [r + (1-\delta)]S_1 + Y_2 \end{cases} \text{,where we assume } \begin{cases} S_0 = S_2 = 0 \\ C_1, C_2 > 0 \end{cases} \end{aligned} \tag{118}$$

$$\text{The reconstraints} \Rightarrow C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r} \text{ (intertemporal budget conconstraint)} \tag{119}$$

Question 1 (How to solve the intertemporal UMP?)**.**

(1). method of elimination

$\frac{du_1}{dC_1} = 0 \Rightarrow \frac{\beta u'(C_2)}{u'(C_1)} = \frac{1}{1+r}$ (the Euler's equation of intertemporal consumption) In which the $\frac{1}{1+r}$ is the comparative price

(2). method of geomotry

On a $C_1 - C_2$ plane,

$$0 = u'(C_1)dC_1 + \beta u'(C_2)dC_2 \tag{120}$$

$$\Rightarrow \frac{dC_2}{dC_1} = -\frac{u'(C_1)}{\beta u'(C_2)} \tag{121}$$

(3). method of Lagrangian function

$$\mathscr{L} = u(C_1) + \beta u(C_2) + \lambda[C_1 + \frac{C_2}{1+r} - Y_1 - \frac{Y_2}{1+r}] \tag{122}$$

$$\Rightarrow \begin{cases} \frac{\partial L}{\partial C_1} = 0 \\ \frac{\partial L}{\partial C_2} = 0 \\ \frac{\partial L}{\partial \lambda} = 0 \end{cases} \tag{123}$$

$$\Rightarrow \frac{\beta u'(C_2)}{u'(C_1)} = \frac{1}{1+r} \text{ (in the difference equation lies the core of Ramsey model)} \tag{124}$$

t-period dynamic optimization (finite-period)

1) The original version:

$$\begin{aligned} &\max_{C_1,C_2,...,C(T)} \; u_1 = u(C_1) + \beta u(C_2) + \beta^2 u(C_3) + ... + \beta^{T-1} u(C_T) \\ &s.t. \begin{cases} C_1 + S_1 & \leqslant [1+r]S_0 + Y_1 \\ C_2 + S_2 & \leqslant [1+r]S_1 + Y_2 \\ & \dots \\ C_T + s_T & \leqslant (1+r)S_{T-1} + Y_T \end{cases} \text{,where we assume} \begin{cases} \delta = s_0 = s_T = 0 \\ C_1, C_2, ..., C_T > 0 \end{cases} \end{aligned} \tag{125}$$

where the budget constraints can be condensed into

$$C_1 + \frac{C_2}{1+r} + ... + \frac{C_T}{(1+r)^{T-1}} = Y_1 + \frac{Y_2}{1+r} + ... + \frac{Y_T}{(1+r)^{T-1}} \tag{126}$$

2) Or the second version:

$$\begin{aligned} &\max_{\{C_t\}_{t=1}^{\infty}} \; u_1 = \sum_{t=1}^{T} \beta^{t-1} u(C_t) \\ &s.t. \begin{cases} C_1 + S_1 & \leqslant [1+r]S_0 + Y_1 \\ C_2 + S_2 & \leqslant [1+r]S_1 + Y_2 \\ & \dots \\ C_T + S_T & \leqslant (1+r)S_{T-1} + Y_T \end{cases} \text{,where we assume} \begin{cases} S_0 = S_T = 0, \delta = 0 \\ C_1, C_2, ..., C_T > 0 \end{cases} \end{aligned} \tag{127}$$

To solve this t-period UMP, using Langranian method: (the λ is the shadow price)

$$\max_{\{C_t\}_{t=1}^{\infty}, \lambda} \; \mathscr{L} = \sum_{t=1}^{T} \beta^{t-1} u(C_t) + \lambda [c_1 + \frac{c_2}{1+r} + ... + \frac{C_T}{(1+r)^{T-1}} - (Y_1 + \frac{Y_2}{1+r} + ... + \frac{Y_T}{(1+r)^{T-1}})] \tag{128}$$

$$\Rightarrow F.O.C.s \begin{cases} \frac{\partial \mathscr{L}}{\partial c_1} = 0 \\ \frac{\partial \mathscr{L}}{\partial C_2} = 0 \\ \vdots \\ \frac{\partial \mathscr{L}}{\partial C_T} = 0 \end{cases} \tag{129}$$

$$\Rightarrow \frac{\beta u'(C_{t+1})}{u'(C_t)} = \frac{1}{1+r}, t = 1, 2 \dots, T \tag{130}$$

3) Another commonly adopted simplified version is

$$\begin{aligned} &\max_{\{C_t, S_t\}_{t=1}^{T}} \; u_1 = \sum_{t=1}^{T} \beta^{T-1} u(C_t) \\ &s.t. C_t + S_t \leqslant (1+r)S_{t-1} + Y_t, \text{ where} \begin{cases} C_t > 0 \\ S_0 = S_T = 0 \end{cases} \end{aligned} \tag{131}$$

Using Lagrangian function method:

$$\Rightarrow u'(C_{t+1}) = \lambda_{t+1} \tag{132}$$

$$\Rightarrow \frac{u'(C_{t+1})}{u'(C_t)} = \frac{\lambda_{t+1}}{\lambda_t} = \frac{1}{\beta(1+r)} \tag{133}$$

$$\Rightarrow \frac{\beta u'(C_{t+1})}{u'(C_t)} = \frac{1}{1+r} \tag{134}$$

infinite-period dynamic optimization
Using the third version, we delve into infinity

$$\begin{aligned} &\max_{\{C_t\}_{t=1}^{\infty}} \; u_1 = \sum_{t=1}^{\infty} \beta^{T-1} u(C_t) \\ &s.t. C_t + S_t \leqslant (1+r) S_{t-1} + Y_t, \text{ where } \begin{cases} C_t > 0 \\ S_0 = \lim\limits_{t\to\infty} S_T = 0 \end{cases} \end{aligned} \tag{135}$$

Using Lagrangian function method:

$$\mathscr{L} \equiv \sum_{t=0}^{\infty} \{\beta^t u(c_t) + \lambda_t [(1+r) s_{t-1} + Y_t - (c_t + s_t)]\} \tag{136}$$

$$\Rightarrow F.O.C. \tag{137}$$

$$\Rightarrow u'(c_{t+1}) = \lambda_{t+1} \tag{138}$$

$$\Rightarrow \frac{u'(c_{t+1})}{u'(c_t)} = \frac{\lambda_{t+1}}{\lambda_t} = \frac{1}{\beta(1+r)} \tag{139}$$

$$\Rightarrow \frac{\beta u'(c_{t+1})}{u'(t)} = \frac{1}{1+r} \tag{140}$$

4.2.3 Method of Hamiltonian Equation (Optimal Control)

Method of Hamiltonian equation is part of the optimal control theory from mathematicals.
a) discrete time

$$\begin{aligned} &\max_{\{c_t, s_t\}_{t=0}^{T}} \; u_1 = \sum_{t=1}^{T} \beta^{T-1} u(c_t) \\ &s.t. c_t + s_t \leqslant (1+r) s_{t-1} + Y_t, \text{ where } \begin{cases} c_t > 0 \\ s_{-1} = s_T = 0 \\ \beta = \frac{1}{1+\rho} \end{cases} \end{aligned} \tag{141}$$

b) continuous time without patience

$$\begin{aligned} &\max_{\{C(t), S(t)\}} \; u = \int_{t=0}^{T} e^{-\rho t} u[C(t)] dt = \int_{t=0}^{T} u[C(t)] dt \\ &s.t. \; C(t) + \dot{S}(t) = r \cdot S(t) + Y(t), \text{ where } S(0) = S(T) = 0 \end{aligned} \tag{142}$$

Using Lagrangian

$$\Rightarrow \mathscr{L} = \int_{t=0}^{T} \{u[C(t)] + \lambda(t)[C(t) + \dot{S}(t) - r \cdot S(t) - Y(t)]\} dt \tag{143}$$

$$= \int_{t=0}^{T} \{H(t) - \lambda(t)\dot{S}(t)\} dt \ (H \text{ is Hamiltonian}) \tag{144}$$

$$= \int_{t=0}^{T} H(t)dt - \int_{t=0}^{T} \lambda(t)\dot{S}(t)dt \ (\text{Then applying fractional integral method to latter}) \tag{145}$$

$$= \int_{t=0}^{T} H(t)dt - \{\lambda(t)S(t)|_0^T - \int_{t=0}^{T} \dot{\lambda}(t)S(t)dt\} \tag{146}$$

$$= \int_{t=0}^{T} H(t)dt - \dot{\lambda}(t)S(t)dt - [\lambda(t)S(t)]|_0^T \tag{147}$$

$$= \int_{t=0}^{T} H(t)dt + \dot{\lambda}(t)dt \tag{148}$$

with this simplified version containing Hamiltonian

$$\Rightarrow \text{F.O.C. of Lagrangian function: } \begin{cases} \frac{\partial \mathscr{L}}{\partial C(t)} = 0 \ (C(t) \text{ is the control variable}) \\ \frac{\partial \mathscr{L}}{\partial S(t)} = 0 \ (S(t) \text{ is the state variable}) \\ \frac{\partial \mathscr{L}}{\partial \lambda(t)} = 0 \ (\lambda(t) \text{ is the monodromy variable}) \end{cases} \tag{149}$$

$$\Rightarrow \text{the optimal control } \begin{cases} \frac{\partial \mathscr{H}}{\partial C(t)} = 0 \\ \frac{\partial \mathscr{H}}{\partial S(t)} = -\dot{\lambda}(t) \\ \frac{\partial \mathscr{H}}{\partial \lambda(t)} = \dot{S}(t) \end{cases} \tag{150}$$

Note 2 (control variable and state variable)**.** *If a variable is determined in the current period, it is a control variable. If it is determined period(s) before, it is a state variable.*

c) continuous time with patience

Next we consider the Lagrangian function with subjective discount rate

Right into this version's Lagrangian function

$$\mathscr{L} = \int_{t=0}^{T} \{e^{-\rho t}u[C(t)] + \lambda(t)[C(t) + \dot{S}(t) - r \cdot S(t) - Y(t)]\} dt \tag{151}$$

$$= \int_{t=0}^{T} \{e^{-\rho t}u[C(t)] + \lambda(t)[rS(t) + Y(t) - C(t)] - \lambda\dot{S}(t)\} dt \tag{152}$$

$$= \int_{t=0}^{T} \{H(0) - \lambda\dot{S}(t)\} dt \ (H(0) \text{ is present value Hamiltonian}) \tag{153}$$

$$\Rightarrow H(0) \equiv e^{-\rho t}u[C(t)] + \lambda(t)[rS(t) + Y(t) - C(t)] \tag{154}$$

$$\Rightarrow \tilde{H}(t) = e^{\rho t}H(0) \ (\text{the current value Hamiltonian}) \tag{155}$$

$$\Rightarrow H(t) = u[C(t)] + \eta(t)[rS(t) + Y(t) - C(t)] \tag{156}$$

Comparing the two scenarios (with and without consideration for ρ), we can see — regardless of the subjective discount rate, result would be the same.

4.2.4 Method of Bellman Equation (Dynamic Programming)

a) discrete time

UMP as below

$$\begin{aligned} &\max_{\{C_t,S_t\}_{t=0}^{\infty}} u_0 = \beta^t u(C_t) \\ &s.t.\ C_t + S_t = (1+r)S_{t-1} + Y_t, \text{ where } S_1 = \lim_{t\to\infty} s_t = 0 \end{aligned} \tag{157}$$

Using the budget constraint to rewrite utility function

$$\Rightarrow C_t = (1+r)S_{t-1} + Y_t - S_t \tag{158}$$

$$\Rightarrow u_0 = \max_{\{S_t\}_{t=0}^{\infty}} \beta^t u\big[(1+r)S_{t-1} + Y_t - S_t\big] \tag{159}$$

$$= \max_{S_0} \beta^t u\big[(1+r)S_{-1} + Y_0 - S_0\big] + \beta \max_{\{S_t\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} u\big[(1+r)S_{t-1} + Y_t - S_t\big] \tag{160}$$

$$= \max_{s_0} u\big[(1+r)S_{-1} + Y_0 - S_0\big] + \beta u(S_0, Y_1) \tag{161}$$

Upscript o denotes optimal

$$\Rightarrow u^o(S_{t-1}, Y_t) = \max_{S_t}\{u\big[(1+r)S_{t-1} + Y_t - S_t\big] + \beta u^o(S_t, Y_{t+1})\} \text{ (Bellman equation)} \tag{162}$$

$$= \max_{S_t}\{u(C_t) + \beta u^o(S_t, Y_{t+1})\} \tag{163}$$

$$\Rightarrow \text{F.O.C.} : u_{C_t}\frac{\partial C_t}{\partial S_t} + \beta\frac{du(S_t, Y_{t+1})}{dS_t} = 0 \tag{164}$$

Let $\frac{\partial C_t}{\partial S_t}$ be -1

$$\Rightarrow \beta\frac{du(s_t, Y_{t+1})}{ds_t} = u_{c_t} \tag{165}$$

Move one period ahead and acquire Euler's equation

(166)

$$u(s_t, Y_{t+1}) = \max_{s_{t+1}}\{u\big[(1+r)s_t + Y_t - s_{t+1}\big] + \beta u(s_{t+1}, Y_{t+2})\} \tag{167}$$

$$\Rightarrow \frac{du^o(s_t, Y_{t+1})}{ds_t} = \frac{\partial u(\cdot)}{\partial s_t} + \frac{\partial u(\cdot)}{\partial s_{t+1}}\frac{ds_{t+1}}{ds_t} + \beta\frac{du^o(\cdot)}{ds_t}\frac{ds_{t+1}}{ds_t} \tag{168}$$

$$= \frac{\partial u(\cdot)}{\partial s_t} + \big[\frac{\partial u(\cdot)}{\partial s_{t+1}} + \beta\frac{du^o(\cdot)}{d_{s_{t+1}}}\big]\frac{ds_{t+1}}{ds_t} \text{ (using optimals' envelope therome)} \tag{169}$$

$$= u_{c,t+1}\frac{dc_{t+1}}{ds_t} + 0 \tag{170}$$

$$= u_{c,t+1}\cdot(1+r) \tag{171}$$

$$\Rightarrow \beta\frac{u_{c,t+1}}{u_{c,t}} = \frac{1}{1+r} \tag{172}$$

Hence the same result as before.

b) continuous time (Hamilton-Jacob-Bellman, HJB)

See in ...

4.3 Intertemporal Equilibrium of Ramsey model

4.3.1 Dynamics of c

Recalling that

$$\frac{\dot{c(t)}}{c(t)} = \frac{r(t) - \rho - \theta g}{\theta} \tag{173}$$

Since we know

$$r(t) = f'[k(t)] \tag{174}$$

Combining them as

$$\frac{\dot{c(t)}}{c(t)} = \frac{r(t) - \rho - \theta g}{\theta} = \frac{f'[k(t)] - \rho - \theta g}{\theta} \tag{175}$$

Once we assume that

$$\exists k^* \in \mathbb{R}, f'[k^*] = \rho - \theta g \tag{176}$$

The dynamics of c, denoted as $\frac{\dot{c(t)}}{c(t)}$ is purely dependent on the value of k according to Ramsey model.

4.3.2 Dynamics of k

Since in Ramsey model we don't assume depreciation of capital k, denoted as $\delta = 0$. The dynamics of k accordingly is $\dot{k}(t) = f[k(t)] - c(t) - (n + g)k(t)$.

This way, in order for the dynamics of k returning to 0, the model must fulfil the requirement of $c(t) = f[k(t)] - (n + g)k(t)$.

4.3.3 Phase diagram and dynamic efficiency

Combining the the two dynamics together, we would have

$$\dot{k} = 0 \tag{177}$$

$$\dot{c} = 0 \tag{178}$$

in a coordinate system of k and c.

Note 3 (dynamic efficiency in Ramsey model)**.** *Beware that the equilibrium of two dynamics would always be on the left of golden rate of $\dot{k}$. The origin point $(0, 0)$ can also be considered a special case of equilibrium.*

The equilibrium in Ramsey model has given only one equilibrium path.

Recalling the Solow presented as model equation of motion

$$k_{t+1} - (1-\delta)k_t = sF(k_t) \tag{179}$$

RCK model has two equations of motion

$$\begin{cases} \frac{u'(c_t)}{u'(c_{t+1})} = \beta[F'(k_{t+1}) - (1-\delta)] \\ k_{t+1} - (1 - \delta k_t) = F(k_t) - C_t \end{cases} \tag{180}$$

Acquirring the steady state of Ramsey model

$$\begin{cases} k_{t+1} = k_t = k^*, \triangle k = 0 \\ c_{t+1} = c_t = c^*, \triangle c = 0 \end{cases} \tag{181}$$

$$\Rightarrow F'(k^*) = \rho + \delta \Rightarrow k^* \tag{182}$$

$$\Rightarrow c^* = F(k^*) - \delta k^* \tag{183}$$

$$\Rightarrow (k^*, c^*) \text{ is the steady state} \tag{184}$$

$$\Rightarrow \frac{dc^*}{dk^*} = F'(k) - \delta = R - \delta \tag{185}$$

Deriving the derivatives at steady state

$$\begin{cases} \frac{\partial \triangle c}{\partial k} < 0 \\ \frac{\partial \triangle k}{\partial c} < 0 \end{cases} \tag{186}$$

, meaning there must be seperating line for each

$$\triangle c = 0 \tag{187}$$

$$\Rightarrow F'(k^*) = \rho + \delta \text{ (from the first equation of motion of RCK)} \tag{188}$$

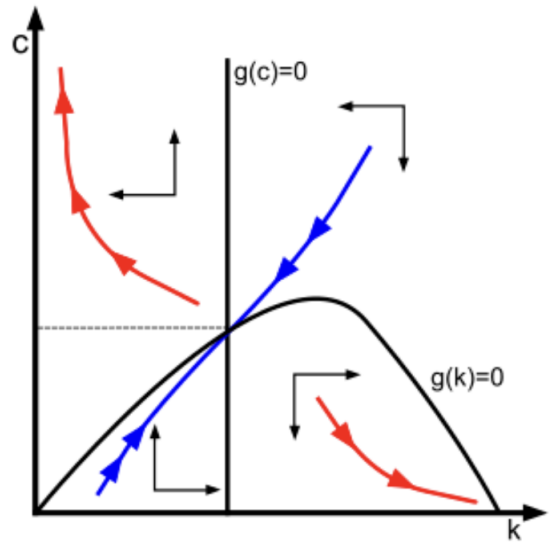
$$\Rightarrow F_{kk}dk = 0dc \tag{189}$$

$$\Rightarrow \frac{dc}{dk} = \frac{F_{kk}}{0} \Rightarrow \text{the seperating line is vertical in C-K plane} \tag{190}$$

$$\triangle k = 0 \tag{191}$$

$$\Rightarrow C^* = F(k^*) - \delta k^* \tag{192}$$

$$\Rightarrow \frac{dc}{dk} = F_k - \delta \Rightarrow \text{the seperating line is vertically reversed U shape} \tag{193}$$



4.4 Linearization

Comparing Solow and Ramsey:
Solow model's Euler equation

$$\dot{k}(t) + \delta k(t) = sF(k(t)) \tag{194}$$

Ramsey model's Euler equations

$$\dot{c}(t) = \frac{F'[k(t) - (\delta + \rho)]}{\theta} c(t), \text{ where } \theta = -\frac{cu''(c)}{u'(c)} \tag{195}$$

$$\dot{k}(t) = F[k(t)] - \delta k(t) - c(t) \tag{196}$$

Noticing the two equations of Ramsey are unlinear. This demands a process of linearization.
Linearization, method of Taylor expansion
First method we use the same technique from Solow model's linearization, Taylor's Expansion.
First order Taylor expansion of RCK' Euler equations:

$$\begin{bmatrix} \dot{c}(t) \\ \dot{k}(t) \end{bmatrix} = \begin{bmatrix} \frac{F'[k(t)-(\delta+\rho)]}{\theta} c(t) \\ F[k(t)] - \delta k(t) - c(t) \end{bmatrix} = \begin{bmatrix} \phi^1[c(t), k(t)] \\ \phi^2[c(t), k(t)] \end{bmatrix} \tag{197}$$

$$\Rightarrow \begin{bmatrix} \dot{c}(t) \\ \dot{k}(t) \end{bmatrix} = \begin{bmatrix} \phi^1[c^*, k^*] \\ \phi^2[c^*, k^*] \end{bmatrix} + \begin{bmatrix} \phi^1_c & \phi^1_k \\ \phi^2_c & \phi^2_k \end{bmatrix} |_{c^*,k^*} \begin{bmatrix} c(t) - c^* \\ k(t) - k^* \end{bmatrix} \tag{198}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} o & \frac{F''(k^*)}{\theta} c^* \\ -1 & \rho \end{bmatrix} \begin{bmatrix} c(t) - c^* \\ k(t) - k^* \end{bmatrix} \tag{199}$$

The above deduction use the following conclusions

$$\begin{cases} F'(k^*) - (\delta + \rho) = 0 \Rightarrow F'(k^*) = \rho + \delta \Rightarrow k^* \\ c^* = F(k^*) - \delta k^* \Rightarrow c^* \end{cases} \tag{200}$$

Because ...

$$Ax = \lambda x \tag{201}$$
$$\Rightarrow \lambda^2 - tr(A)\lambda + dt + \lambda = 0 \tag{202}$$
$$\Rightarrow \lambda^2 - \rho\lambda + \frac{F'(k^*)}{\theta} \cdot c^* = 0 \tag{203}$$
$$\Rightarrow \lambda_1 = \frac{\rho - \sqrt{\rho^2 - 4F''(k^*)c^*}}{\theta} < 0 < \lambda_2 = \frac{\rho + \sqrt{\rho^2 - 4F''(k^*)c^*}}{\theta} \tag{204}$$

$\Rightarrow$ We choose the negative one which is λ_1 so that $k(t)$ can converge over time

$$\Rightarrow \dot{k}(t) = \lambda_1[k(t) - k^*] \tag{205}$$
$$\Rightarrow k(t) = k^* + e^{\lambda_1 t}[k(0) - k^*] \tag{206}$$

Linearization, method of undetermined coefficients
The second approach is method of undetermined coefficients

$$\dot{c}(t) = \frac{F''(k^*)}{\theta}c^* = 0 \tag{207}$$
$$\dot{k}(t) = \rho[k(t) - k^*] - [c(t) - c^*] \tag{208}$$
$$\Rightarrow \ddot{k}(t) = \rho\dot{k}(t) - \dot{c}(t) \tag{209}$$
$$= \rho\dot{k}(t) - \frac{F''(k^*)}{\theta} \cdot c^*[k(t) - k^*] \tag{210}$$

We can make an assumption that $\dot{k}(t) = \lambda[k(t) - k^*]$, which is a differential equation in nature

$$\Rightarrow \ddot{k}(t) = \lambda\dot{k}(t) \tag{211}$$
$$= \lambda^2[k(t) - k^*] \tag{212}$$
$$= \rho\dot{k}(t) - \frac{F''(k^*)c^*[k(t) - k^*]}{\theta} \tag{213}$$
$$= \lambda^2 - \rho\lambda + \frac{F''(k^*)c^*[k(t) - k^*]}{\theta} = 0 \tag{214}$$
$$\Rightarrow \lambda_1 = \frac{\rho - \sqrt{\rho^2 - 4F''(k^*)c^*}}{\theta} < 0 < \lambda_2 = \frac{\rho + \sqrt{\rho^2 - 4F''(k^*)c^*}}{\theta} \tag{215}$$

Samewise

$$\Rightarrow k(t) = k^* + e^{\lambda_1 t}[k(0) - k^*] \tag{216}$$

Note 4 (convergence in Ramsey model)**.** *Recall Solow model's chapter of convergence speed and time, we can compare the two similar results:*

$$\begin{cases} \textit{Ramsey: } k(t) = k^* + e^{\lambda_1 t}[k(0) - k^*] \\ \textit{Solow: } k(t) = k^* + e^{[g'(k^*)t]}[k(0) - k^*] \end{cases} \tag{217}$$

Meaning the λ we are trying to derive here is the convergence speed.

4.5 Maximizing welfare

Similiar to the concept of golden rate in Solow model, Ramsey model also provide a glimpse of maximized consumption.

At the steady state

$$c^* = F(k^*) - \delta k^* \tag{218}$$

$$\Rightarrow k^*_{gold} = \underset{k^*}{\text{argmax}}\ c^* \tag{219}$$

4.6 Taxation and capital accumulation

Planned section. Compare lump-sum and distortionary taxes in the neoclassical growth framework. Separate effects on steady-state capital from effects along the transition.

5 OLG Model

In contrast to the Ramsey model in which individuals are infinitely-lived, in the OLG model individuals live a finite length of time, long enough to overlap with at least one period of another agent's life.

Note 5 (Meet only once)**.** *Two generation OLG is particularly simple because it precludes intertemporal trade (no one meets twice).*

Solow model, RCK model and OLG model

- All of them are under the assumption of perfect competitive market.
- Solow model has exogenous saving rate, whereas Ramsey model and OLG model has endogenous one.
- OLG model has heterogeneous representative agents.

5.1 Assumptions

$$C_t = L_t C_{1t}^{\text{young}} + L_{t-1} C_{2t}^{\text{old}} \text{ (where subscript 1 and 2 stands for age)} \tag{220}$$

$$L_t = (1+n)L_{t-1} \Rightarrow L_t = (1+n)^t L_0 \tag{221}$$

$$S_t = s_t L(t) \tag{222}$$

5.2 UMP of household

A general UMP would be

$$\max u(C_1) + \beta u(C_{2,t+1}) \quad s.t. \begin{cases} C_{1t} + S_t \leqslant W_t \\ C_{2,t+1} \leqslant R_{t+1} S_t \end{cases} \tag{223}$$

Form a Lagrangian and acquire Euler's equation

$$\mathscr{L} = u(C_1) + \beta u(C_{2,t+1}) + \lambda_{1t}(c_{1t} + s_t - w_t) + \lambda_{2,t+1}(c_{2,t+1} - R_{t+1} s_t) \tag{224}$$

$$\Rightarrow \text{F.O.C.} \begin{cases} \frac{\partial \mathscr{L}}{\partial C_{1t}} = 0 \\ \frac{\partial \mathscr{L}}{\partial C_{2,t+1}} = 0 \\ \frac{\partial \mathscr{L}}{\partial S_t} = 0 \end{cases} \Rightarrow \begin{cases} u'(C_{1t}) = \lambda_{1t} \\ \beta' u'(C_{2,t+1}) = \lambda_{2,t+1} \\ \frac{\lambda_{1t}}{\lambda_{2,t+1}} = R_{t+1} \end{cases} \tag{225}$$

$$\Rightarrow u'(C_{1t}) = \beta R_{t+1} u'(C_{2,t+1}) \text{ (Euler's equation in OLG UMP)} \tag{226}$$

e.g. Given a specific utillity function

$$u(C_{gt}) = \frac{C_{gt}^{1-\theta} - 1}{1-\theta}, 0 < \theta \neq 1, g = 1, 2 \tag{227}$$

$$\text{UMP} \Rightarrow \begin{cases} C_{1t}^{-\theta} = \beta R_{t+1} C_{2,t+1}^{-\theta} \\ C_{1t} = W_t - S_t \\ C_{2,t+1} = R_{t+1} S_t \end{cases} \tag{228}$$

$$\Rightarrow (W_t - S_t)^{-\theta} = \beta R_{t+1} (R_{t+1} S_t)^{-\theta} \tag{229}$$

$$\Rightarrow (\frac{W_t - S_t}{S_t})^{-\theta} = \beta R_{t+1}^{1-\theta} \tag{230}$$

$$\Rightarrow \frac{W_t}{S_t} = (\beta R_{t+1}^{1-\theta})^{-\frac{1}{\theta}} + 1 \tag{231}$$

$$\Rightarrow S_t = \frac{w_t}{\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1} < w_t \tag{232}$$

$$\Rightarrow S'_w = \frac{1}{\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1} \in (0, 1) \tag{233}$$

$$\Rightarrow S'_R = \frac{1-\theta}{\theta} \frac{S_t}{\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1} (\beta R_{t+1})^{-\frac{1}{\theta}} \tag{234}$$

$$\Rightarrow S'_R \begin{cases} > 0, 0 < \theta < 1 \\ = 0, \theta = 1 \\ < 0, \theta > 1 \end{cases} \tag{235}$$

5.3 PMP of firm

A general PMP would be

$$\max_{k_t, L_t} \pi_t = Y_t - w_t L_t - R_t K_t \tag{236}$$

$$\text{Since } \pi_t = \frac{\Pi_t}{L_t}, y_t = \frac{Y_t}{L_t}, k_t = \frac{K_t}{L_t}, \frac{K_{t+1}}{L_t} = \frac{K_{t+1}}{L_{t+1}} \frac{L_{t+1}}{L_t}, s_t = \frac{S_t}{L_t} \tag{237}$$

Rearrange in form of per capita

$$\Rightarrow \max_{K_t} \pi_t = Y_t - W_t - R_t K_t \tag{238}$$

$$\Rightarrow \text{F.O.C.} \begin{cases} R_t = f'(K_t) \\ W_t = f(K_t) - K_t f'(K_t) \end{cases} \tag{239}$$

5.4 Dynamics

Looking at the motion equation of capital

$$K_{t+1} - (1-\delta)K_t = I_t = S_t = s_t L_t = s(w_t, R_{t+1})L_t \tag{240}$$

$$\Rightarrow K_{t+1} = \frac{s(w_t, R_{t+1})}{1+n} \tag{241}$$

$$\Rightarrow K_{t+1} = \frac{w_t}{(\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1)(1+n)} \text{ (using conditions from UMP and PMP)} \tag{242}$$

$$\Rightarrow K_{t+1} = \frac{f(K_t) - K_t f'(K_t)}{(\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1)(1+n)} \text{ (equation of motion of capital in OLG)} \tag{243}$$

The steady states (intertemporal equilibrium) would be

$$k_{t+1} = k_t = k^* \tag{244}$$

$$\Rightarrow k^* = \frac{s[f(k_t^*) - k_t^* f'(k_t^*), f'(k^*)]}{1+n} \tag{245}$$

$$= \frac{f(k_t^*) - k_t^* f'(k_t^*)}{(\beta^{-\frac{1}{\theta}} f'(*)^{-\frac{1-\theta}{\theta}} + 1)(1+n)} \tag{246}$$

$$\Rightarrow k_{t+1} = \frac{(1-\alpha)(k^*)^\alpha}{(\beta^{-\frac{1}{\theta}}[\alpha(k^*)^{\alpha-1}]^{-\frac{1-\theta}{\theta}} + 1)(1+n)} \text{ (capital per capita at steady state in OLG)} \tag{247}$$

$$\Rightarrow k^* = \cdot \text{ (thus we have the } k \text{ at steady state)} \tag{248}$$

e.g., For a utillity function $u = \log c_{1t} + \beta \log c_{2,t+1}$, where $\theta = 1$.

$$\Rightarrow \frac{c_{2,t+1}}{c_{1t}} = \beta k_{t+1} \tag{249}$$

$$\Rightarrow s_t = \frac{w_t}{\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1} \tag{250}$$

$$= \frac{w_t}{\beta^{-1} + 1} \tag{251}$$

$$= \frac{\beta}{1+\beta} w_t \text{ (meaning the saving is a function of wage, instead of endowment)} \tag{252}$$

$$\Rightarrow k_{t+1} = \frac{s_t}{1+n} = \frac{\beta w_t}{(1+\beta)(1+n)} \tag{253}$$

$$= \frac{\beta(1-\alpha k_t^\alpha)}{(1+\beta)(1+n)} \tag{254}$$

$$\text{steady state demands } k_{t+1} = k_t = k^* \Rightarrow k^* = \left[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}\right]^{\frac{1}{1-\alpha}} \tag{255}$$

5.5 Comparing three models so far

$$\text{Solow}\begin{cases} L_t = (1+n)L_{t-1} \\ \frac{S_t}{L_t} = s_t = sy_t = s\frac{Y_t}{L_t} \\ k_{t+1} = \frac{(1-\delta)k_t + sf(k_t)}{1+n} \end{cases} \tag{256}$$

$$\text{OLG}\begin{cases} L_t = (1+n)L_{t-1} \\ k_{t+1} = \frac{s(w_t, R_{t+1})}{1+n} = \frac{s[f(k_t) - k_t f'(k_t), f'(k_{t+1})]}{1+n} \end{cases} \tag{257}$$

$$\text{Ramsey}\begin{cases} L_t = (1+n)L_{t-1} \\ k_{t+1} = \frac{f(k_t)+c_t}{1+n} \Rightarrow 0 = F(k^*) - c^* - \rho k^* \Rightarrow F'(k_G^*) = \delta \\ \frac{u'(c_t)}{u'(c_{t+1})} = \beta[f'(k_{t+1}) + (1-\delta)] \Rightarrow 1 = \rho[F'(k^*) + (1-\delta)] \Rightarrow F'(k^*) = \rho + \delta \Rightarrow k_G^* > k^* \end{cases} \tag{258}$$

5.6 Dynamic inefficiency

Because of the micro-foundations (patience denoted as $\beta = \frac{1}{1+\rho}$) in Ramsey, there will be no over-accumulation of capital, avoiding dynamic inefficiency. Whereas in Solow model the inefficiency is inevitable. Somehow in OLG model, the dynamic inefficiency returns.

Solow

The model has key equations

$$\begin{cases} Y = F(K, L) \\ I = S = sY \\ L_{t+1} = (1+n)L_t \\ k_{t+1} = \frac{(1-\delta)k_t + sf(k_t)}{(1+n)} \iff \dot{k} = sf(k) - (n+\delta)k \end{cases} \tag{259}$$

For a certain production $y = k^a$ at steady state

use $k_{t+1} = \frac{(1-\delta)k_t + sf(k_t)}{1+n}$

$$\Rightarrow (1+n)k^* = (1-\delta)k^* + sf(k^*) \tag{260}$$

$$-(n+\delta)k^* = -sf(k^*) \tag{261}$$

$$f(k^*) - (n+\delta)k^* = (1-s)f(k^*) = c^* \tag{262}$$

$$\Rightarrow c^* = f(k^*) - (n+\delta)k^* \tag{263}$$

$$\text{To maximize welfare at steady state} \iff \max c* \iff FOC : k_g^* = (\frac{n+\delta}{\alpha})^{1-\alpha} \tag{264}$$

also at steady state

$$(1+n)k^* = (1-\delta)k^* + sf(k^*) \tag{265}$$

$$\Rightarrow k^* = (\frac{n+\delta}{s})^{1-\alpha} \tag{266}$$

for certain α and s it could be

$$\Rightarrow k^* < k_g^* \tag{267}$$

meaning there is possible dynamic inefficiency in Solow model

Ramsey

Ramsey has no dynamic inefficiency, here's why

The model has three key equations

$$\begin{cases} L_t = (1+n)L_{t-1} \\ k_{t+1} - (1-\delta)k_t = f(k_t) - c_t \\ \frac{u'(c_t)}{u'(c_{t+1})} = \beta[f'(k_{t+1}) + (1-\delta)] \end{cases} \tag{268}$$

Put $k_{t+1} - (1-\delta)k_t = f(k_t) - c_t$ at steady state

$$c^* = f(k^*) - k^* + (1-\delta)k^* = f(k^*) - \delta k^* \tag{269}$$

$$\text{To maxmize welfare at steady state} \Rightarrow \text{FOC: } f'(k_G^*) = \delta \tag{270}$$

Put $\frac{u'(c_t)}{u'(c_{t+1})} = \beta[f'(k_{t+1}) + (1-\delta)]$ at steady state resulting

$$\Rightarrow 1 = \beta[f'(k^*) + (1-\delta)] = \frac{1}{1+\rho}[f'(k^*) + (1-\delta)] \tag{271}$$

$$\Rightarrow 1 + \rho = f'(k^*) + 1 - \delta \tag{272}$$

$$\Rightarrow f'(k^*) = \rho + \delta < f'(k_g^*) \tag{273}$$

$$\Rightarrow k_G^* > k^* \tag{274}$$

Capital per capita at steady state is lower than k_{gold}, meaning it is dynamic efficient.

OLG

OLG in central planner form:

$$\begin{aligned} &\max \sum_{t=0}^{\infty} (\beta_s)^t L_t[u(c_{1t}) + \beta u(c_{2,t+1})] \text{ (β_s is the discount factor)} \\ &s.t.\ c_{1t} + \frac{c_{2t}}{1+n} + (1+n)k_{t+1} = f(k_t) \end{aligned} \tag{275}$$

Equation of motion of capital per capita at the steady states

$$c^* \equiv c_1^* + \frac{c_2^*}{1+n} = f(k^*) - (1+n)k^* \tag{276}$$

Since we are maximizing welfare at steady state, let $\frac{\partial c^*}{\partial k^*} = 0$ to acquire first order condition

$$\Rightarrow f'(k_G^*) = 1 + n \tag{277}$$

using production function $y = k^\alpha$ at equilibrium

$$f'(k^*) = \alpha(k^*)^{\alpha-1} \tag{278}$$

$$= \alpha\{[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}]^{\frac{1}{1-\alpha}}\}^{\alpha-1} \tag{279}$$

$$= \frac{\alpha}{1-\alpha}\frac{1+\beta}{\beta}(1+n) \tag{280}$$

for certain value of α

$$f'(k^*) < 1 + n \tag{281}$$
$$\Rightarrow f'(k^*) < f'(k_G^*) \tag{282}$$
$$\Rightarrow k^* > k_G^* \tag{283}$$

Meaning there can be over-accumulation of capital $\iff$ suffices dynamic inefficiency possible in OLG model.

Conclusion

- Due to ad hoc given s, Solow model is unable to assure dynamic efficiency.
- Due to endogenous saving decision and discount factor β, Ramsey model is able to achieve dynamic efficiency.
- Due to limited life span of individuals, OLG model fails to achieve dynamic efficiency.

5.7 Population, pensions, and public debt

Planned section. Extend the two-generation benchmark to population change, pension arrangements, and government debt. State the conditions for dynamic inefficiency before comparing intergenerational transfers.

6 Endogenous Growth and Resource Allocation

Planned chapter.

6.1 The AK benchmark and human capital

Ask when reproducible factors can sustain growth without exogenous technological progress. Compare the roles of returns to capital and human-capital accumulation.

6.2 Ideas, innovation, and incentives

Introduce nonrival ideas and incentives to innovate. Separate the private return to innovation from its social return and discuss the policy implications.

6.3 Allocation and aggregate productivity

Explain how differences in productivity and financing conditions affect the allocation of capital. Connect allocation to aggregate output and growth.

Part III
Fluctuations

How do shocks change production, spending, employment, and prices? We begin with a real benchmark, introduce nominal rigidities, and then examine labor-market adjustment. Part V adds financial and distributional mechanisms and develops the policy comparison.

7 Real Business Cycle Model

Originated from Kyland and Prescott (1982), Long and Plosser (1983), RBC model shows great empirical importance. A standard RBC model[11] is in nature a Ramsey model with

- Endogenous (but constant in the long run, i.e. $n = 0$) labor supply
- Stochastic productivity (i.e. real shocks)

Note 6 (Two principles of the RBC theory of business cycles)**.** *1. Money is of little importance in business cycles.*

2. Business cycles are created by rational agents responding optimally to real (not nominal) shocks - mostly fluctuations in productivity growth, but also fluctuations in government purchases, import prices, or preferences.

7.1 UMP of household

A typical household entails product-consuming for C_t and labor-providing for N_t. Only in this model, the agent value leisure $L_t = 1 - N_t$.

The UMP in RBC model is

$$\max_{C_t, N_t, B_t} E_t \sum_{t=0}^{\infty} \beta^t u(C_t, N_t) \tag{284}$$

$$s.t.\ P_t C_t + Q_t B_t \leqslant B_{t-1} + W_t N_t + D_t, \text{ where } \begin{cases} \beta \in (0,1) \\ Q_t = \frac{1}{1+i_t} \text{ is the price per bond bought today} \end{cases} \tag{285}$$

Form a Lagrangian and acquire FOCs

$$\mathscr{L} = E_0 \sum_{t=0}^{\infty} \beta^t [u(C_t, N_t) + \lambda_t (P_t C_t + Q_t B_t - B_{t-1} - W_t N_t - D_t)] \tag{286}$$

$$\Rightarrow \begin{cases} \frac{\partial \mathscr{L}}{\partial C_t} = 0 \Rightarrow \beta^t [u_{C_t} - \lambda_t P_t] = 0 \Rightarrow \frac{u_{C_t}}{E_t U_{c,t+1}} = \frac{\lambda_t}{E_t \lambda_{t+1}} \cdot \frac{P_t}{E_t P_{t+1}} \\ \frac{\partial \mathscr{L}}{\partial N_t} = 0 \Rightarrow \beta^t [u_{N_t} + \lambda_t W_t] = 0 \\ \frac{\partial \mathscr{L}}{\partial B_t} = 0 \Rightarrow -\lambda_t Q_t \beta^t + E_t \lambda_{t+1} \beta^{t+1} = 0 \Rightarrow \beta \frac{E_t \lambda_{t+1}}{\lambda_t} = Q_t \Rightarrow \frac{1}{Q_t} = 1 + i_t \end{cases} \tag{287}$$

using these FOCs

$$\begin{cases} \beta^t [u_{C_t} - \lambda_t P_t] = 0 \\ \beta^t [u_{N_t} + \lambda_t W_t] = 0 \end{cases} \Rightarrow \text{Labor supply equation} - \frac{u_{N_t}}{u_{C_t}} = \frac{W_t}{P_t} \tag{288}$$

$$\text{Also} \begin{cases} \frac{u_{C_t}}{E_t U_{c,t+1}} = \frac{\lambda_t}{E_t \lambda_{t+1}} \cdot \frac{P_t}{E_t P_{t+1}} \\ \beta \frac{E_t \lambda_{t+1}}{\lambda_t} = Q_t \end{cases} \Rightarrow \text{Euler's equation } \frac{Q_t}{\beta} = E_t \cdot \frac{\frac{U_{c,t+1}}{P_{t+1}}}{\frac{U_{c,t}}{P_t}} \tag{289}$$

[11] For a sound review over the topic of RBC, see Rebelo Sergio. 2005. "Real Business Cycle Models: Past, Present, and Future." The Scandinavian Journal of Economics.

e.g. Let utility function be specific

$$u(C_t, N_t) = \begin{cases} \frac{c_t^{1-\sigma}-1}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}, \sigma \neq 1 \\ \log C_t - \frac{N_t^{1+\varphi}}{1+\varphi}, \sigma = 1 \end{cases} \tag{290}$$

$$\Rightarrow \begin{cases} \frac{W_t}{P_t} = C_t^{\sigma} N_t^{\varphi} \\ Q_t = \beta E_t \cdot (\frac{C_{t+1}}{C_t})^{-\sigma} \cdot \frac{P_t}{P_{t+1}} \end{cases} \tag{291}$$

7.2 PMP of firm

$$\min_{N_t} W_t N_t \Leftrightarrow \max_{Y_t} P_t Y_t - W_t N_t \tag{292}$$

$$s.t. Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}, \alpha \in [0,1) \tag{293}$$

$$\text{where } \begin{cases} Y_t = A_t W_t^{1-\alpha} \\ A_t = A_{t-1}^{\rho\alpha} \epsilon_t^{\alpha}, \epsilon_t^{\alpha} \overset{iid}{\sim} N(0, \sigma_{\alpha}^{\epsilon}) \text{ (technology shocks)} \end{cases} \tag{294}$$

$$\Rightarrow \text{F.O.C.: } \frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha} \text{ (labor demand equation)} \tag{295}$$

7.3 Economy described by RBC

The whole economy system is described as the balance of demand and supply

$$\text{AS} \begin{cases} Y_t = A_t N_t^{1-\alpha} \text{ (total production function)}, A_t = A_{t-1}^{\rho a} e^{\epsilon_t^a} \text{ (technology shocks)} \\ \frac{W_t}{P_t} = C_t^{\sigma} N_t^{\varphi} \text{ (labor supply from UMP)} \\ \frac{w_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha} \text{ (labor demand from PMP)} \end{cases} \tag{296}$$

$$\text{AD} \begin{cases} Y_t = C_t + I_t \text{ (accouting equation)} \\ Q_t = \beta E_t (\frac{C_{t+1}}{C_t})^{-a} \frac{P_t}{P_{t+1}} \text{ (Euler equation from UMP)} \\ \frac{M_t}{P_t} = \frac{Y_t}{Q_t^{-n}} \text{ (demand of liquidity)} \end{cases} \tag{297}$$

7.4 Log-linearization

Two methods we can transform. 1) The first is

$$\hat{x}_t = \ln x_t - \ln x^* \tag{298}$$
$$= \ln|\frac{x_t}{x^*}| \tag{299}$$
$$= \ln|1 + \frac{x_t - x^*}{x^*}| \text{ (percentage of deviation from stable state)} \tag{300}$$
$$\approx \ln|1 + \frac{1}{x^*}(x_t - x^*)| \tag{301}$$
$$= \frac{x_t - x^*}{x^*} \tag{302}$$

2) Another approach would be

$$\ln x^* = lnx_t - \tilde{x_t} \Rightarrow \ln x_t = lnx^* + \tilde{x_t} \tag{303}$$
$$\Rightarrow e^{\ln x_t} = x_t = e^{nx^* + \tilde{x_t}} = x^* e^{\hat{x_t}} \approx x^*(1 + \hat{x}_t) \tag{304}$$

Now we turn to the linearization of RBC model

$$\begin{cases} y_t = a_t + (1-\alpha)n_t \\ y = a + (1-\alpha)n \end{cases} \Rightarrow \hat{y}_t = \hat{a}_t + (1-\alpha)\hat{n}_t \tag{305}$$

$$\Rightarrow \begin{cases} \hat{w}_t - \hat{p}_t = \hat{a}_t + (1-\alpha)\hat{n}_t \\ \hat{w}_t - \hat{p}_t = a\hat{c}_t + \rho\hat{n}_t \end{cases} \Rightarrow \hat{y}_t = \hat{c}_t,\ m_t - \hat{p}_t = \hat{y}_t - \hat{n}_{it} \tag{306}$$

$$Y_t = C_t + I_t \tag{307}$$
$$Y \cdot e^{\hat{y}_t} = C \cdot e^{\hat{c}_t} + I \cdot e^{\hat{i}_t} \tag{308}$$
$$Y(1 + \hat{y}_t) = C(1 + \hat{c}_t) + I(1 + \hat{i}_t) \tag{309}$$
$$\hat{y}_t = \frac{C}{Y}\hat{c}_t + \frac{I}{Y}\hat{i}_t \tag{310}$$
$$a_t = \rho_t a_{t-1} + \epsilon_t^a \tag{311}$$
$$1 = E_t e^{\ln\beta + + \ln\frac{1}{Q_t} - a(c_{t+1} - c_t) - (P_{t+1} - P_t)} \tag{312}$$
$$1 = E_t e^{\ln\beta + + \ln\frac{1}{Q_t} - a\triangle c_{t+1} - \pi_{t+1}} \tag{313}$$
$$1 = E_t I[1 + (i_t - i) - a(\triangle c_{t+1} - \triangle c_t] - (\pi_{t+1} - \pi_t)] \tag{314}$$

Therefore a RBC model can be rewritten in linearized log equations

$$\text{AS:}\begin{cases}\hat{y}_t = a_t + (1-\alpha)\hat{n}_t \text{ (output)}\\ \hat{w}_t - \hat{p}_t = a_t - \alpha\hat{n}_t \text{ (labor demand)}\\ \hat{w}_t - \hat{p}_t = ac_t + \varphi\hat{n}_t \text{ (labor supply)}\\ \hat{a}_t = \rho_a\hat{a}_{t-1} + \epsilon_t^a \text{ (technology)}\end{cases} \tag{315}$$

$$\text{AD:}\begin{cases}\hat{y}_t = \hat{c}_t \text{ (product market clearing)}\\ \hat{c}_t = E_t\hat{c}_{t+1} - \frac{1}{a}(\hat{i}_t - E_t\hat{\pi}_{t+1})\\ \hat{m}_t - \hat{p}_t = \hat{y}_t - \eta\cdot\hat{i}_t \text{ (money demand equation; } \eta \text{ is interest rate elasticity)}\\ \hat{r}_t = i_t - E_t\hat{\pi}_{t+1} \text{ (Fisherian equation)}\end{cases} \tag{316}$$

Note 7 (Exceptation for Expectation)**.** *Notice this method does not apply to equations involving expectations.*

Note 8 (RBC has neutrality of money)**.** *The final answer of this standard RBC model is not relevant to any price variable, inferring the model being money-neutral.*

7.5 Capital accumulation and investment

Planned section. Extend the current fixed-capital benchmark to endogenous capital accumulation. Derive investment dynamics and distinguish the simplified benchmark from a full RBC model with a changing capital stock.

8 New Keynesian Model

DNK is short for Dynamic New Keynesian[12]
This New keynesian Model contains elements being

- dispersive economy, a continuum of agents
- perfect information
- rational expectations, i.e. $E_t(X_{t+1}) = E(X_{t+1}|I_t)$
- perfect competitive labor market
- monopolisitc competitive product market, heterogenous products, nominal rigidities

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8.1 UMP of household

Two stages of optimization

Goal of UMP is to acquire DIS, which represents demand. Household face two stages of optimization. Since goods are heterogenous, UMP begins by choosing items $i \in I$.

Stage I: choosing goods

One way to choose products is by method of maximization.

$$\begin{aligned} &\max_{C_{it}}(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} \\ &s.t. \int_0^1 P_{it}C_{it}di \leqslant Z_t \end{aligned} \tag{317}$$

, in which the ϵ stands for current substitute elasticity and $\frac{\epsilon-1}{\epsilon}$ is contemporaneous elasticity of substitution. And Z_t stands for endowment.

The other way to choose products is by method of minimization. In this case Lagrangian multiplier could be shadow price.

$$\begin{aligned} &\min \int_0^1 P_{it}C_{it}di \\ &s.t.(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} \geqslant C_t \end{aligned} \tag{318}$$

Form a Lagrangian, we would have

$$\underset{\{C_{it}\}}{\mathscr{L}} = \int_0^1 P_{it}C_{it}di + P_t[(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} - C_t] \tag{319}$$

$$\Rightarrow P_{it} = P_t \frac{\epsilon}{\epsilon-1}((\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}-1}) \frac{\epsilon-1}{\epsilon} C_{it}^{\frac{\epsilon-1}{\epsilon}-1} \tag{320}$$

$$\Rightarrow P_{it} = P_t[(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}})^{\frac{1}{\epsilon}}] C_{it}^{\frac{-1}{\epsilon}} \tag{321}$$

$$\Rightarrow \frac{P_{it}}{P_t} = (\frac{C_{it}}{C_t})^{\frac{-1}{\epsilon}} \tag{322}$$

$$\Rightarrow C_{it} = (\frac{P_{it}}{P_t})^{-\epsilon} C_t \text{ (the demand curve)} \tag{323}$$

[12]This chapter is mainly based on the Gali, 2015.

From $\int_0^1 P_{it}C_{it}di = P_tC_t$ we have

$$\Rightarrow \int_0^1 P_{it}(\frac{P_{it}}{P_t})^{-\epsilon}C_t di = P_tC_t \tag{324}$$

$$\Rightarrow \int_0^1 P_{it}(\frac{P_{it}}{P_t})^{-\epsilon} di = P_t \tag{325}$$

$$\Rightarrow P_t = (\int_0^1 P_{it}^{1-\epsilon}di)^{\frac{1}{1-\epsilon}} \text{ (the expression of total price level)} \tag{326}$$

Apparently $\begin{cases} \max\limits_{C_{it}} (\frac{1}{I}\sum\limits_{i=1}^{I} C_{it}^{\frac{1}{1+n_t}})^{1+n_t} = C_t \\ s.t. \sum\limits_{i=1}^{I} P_{it}C_{it} \leqslant Z_t \end{cases}$ and $\begin{cases} \min\limits_{C_{it}} \sum\limits_{i=1}^{I} P_{it}C_{it} \\ s.t. (\frac{1}{I}\sum\limits_{i=1}^{I} C_{it}^{\frac{1}{1+n_t}})^{1+n_t}) \geqslant C_t \end{cases}$ are the same. But as mentioned before, the minimization approach is commonly prefered.

Using either approach, we can conduct the result that demand curve and expression of total price level being

$$C_{it} = (I \cdot \frac{P_{it}}{P_t})^{\frac{1+n_t}{-\wedge_t}} C_t \tag{327}$$

$$P_t = I(\frac{1}{I}\sum_{i=1}^{I} P_{it}^{\frac{1}{-\wedge_t}})^{-\wedge_t} \tag{328}$$

After picking products, we enter stage II, UMP at period t

$$\begin{aligned} &\max_{C_t, N_t, \frac{M_t}{P_t}} E_0 \sum_{t=0}^{\infty} \beta^t u\left(C_t, N_t, \frac{M_t}{P_t}\right) \leftarrow \text{(MIU)} \\ &s.t. \int_0^1 P_{it}C_{it}di + M_t + B_t \leqslant M_{t-1} + (1+i_{t-1})B_{t-1} + \int_o^1 W_{it}N_{it}di + T_t \end{aligned} \tag{329}$$

constraint of this UMP can be rewritten as

$$\begin{aligned} &\max_{C_t, N_t, \frac{M_t}{P_t}} E_0 \sum_{t=0}^{\infty} \beta^t u\left(C_t, N_t, \frac{M_t}{P_t}\right) \\ &s.t.\ P_tC_t + M_t + B_t \leqslant M_{t-1} + (1+i_{t-1})B_{t-1} + W_tN_t + T_t \end{aligned} \tag{330}$$

Note 9 (Subscript it and t)**.** $C_{it} \neq C_t \Leftrightarrow$ *Monopolistic competitive goods market;* $N_{it} = N_t \Leftrightarrow$ *Perfect competitve labor market*

Form the Lagrangian

$$\mathscr{L} = E_0 \sum_{t=0}^{\infty} \beta^t \{u(C_t, N_t, \frac{M_t}{P_t}) + \lambda_t[P_tC_t + M_t + B_t - M_{t-1} - (1+i_{t-1})B_{t-1} - W_tN_t + T_t]\} \tag{331}$$

Solving this will deliver the F.O.C.s., resulting

$$\Rightarrow \begin{cases} Q_t = \beta E_t \cdot (\frac{u_{C,t+1}}{u_{C,t}} \frac{P_t}{P_{t+1}}) \\ \frac{W_t}{P_t} = -\frac{u_{N_t}}{U_{C_t}} \\ 1 - Q_t = u_{\frac{M_t}{P_t}} \end{cases} \tag{332}$$

This result is basically same as in RBC model with one more equation since money is added to the utility function.

Linearization

Linearization specificly focusing on the third equation of the second equation, as the other two are identical in RBC model's linearization.

$$1 - \frac{1}{1+i_t} = C_t^{\sigma}(\frac{M_t}{P_t})^{-\nu} \text{ (using } Q_t\text{'s defintion)} \tag{333}$$

$$\Rightarrow \frac{i_t}{1+i_t} = C_t^{\sigma}(\frac{M_t}{P_t})^{-\nu} \tag{334}$$

Taking logs

$$\log \frac{i_t}{1+i_t} = \sigma c_t - \nu(m_t - p_t) \text{ (hereafter a lowercase letter } x \text{ denotes } \log x) \tag{335}$$

$$\log \frac{i}{1+i} = \sigma c - \nu(m - p) \tag{336}$$

$$\Rightarrow \log(\frac{i_t}{1+i_t}) - \log(\frac{i}{1+i}) = \sigma \hat{c}_t - \nu(m_t - \hat{p}_t) \tag{337}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + \log(\frac{1+i_t}{i_t}) - \log(\frac{1+i}{i}) \tag{338}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + [\log \frac{(\frac{1+i}{i}) + 1}{\frac{1+i}{i}} \frac{i - (1+i)}{i^2}(i_t - i) - \log(\frac{1+i}{i})] \tag{339}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + \frac{i}{i+1}\frac{i - (1+i)}{i}\frac{i_t - i}{i} \tag{340}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t - \frac{1}{1+i}\hat{i}_t \tag{341}$$

$$\Rightarrow m_t - \hat{p}_t = \frac{\sigma}{\nu}\hat{c}_t - \frac{1}{(1+i)\nu}\hat{i}_t \tag{342}$$

$$\Rightarrow m_t - \hat{p}_t = \hat{c}_t - \eta\hat{i}_t \tag{343}$$

Note 10 (Must-be in exam: monopolistic competition vs perfect competition)**.** *Zero market power to finite market power, meaning infinite elasticiy to non-zero positive elasticity.*

8.2 PMP of firm

Note 11 (Two features DNK firm has)**.** *Monopolistic competitive goods market and stikcy price.*

8.2.1 Price-setting behavioral equation

We first use PMP to acquire price-setting equation then linearize it. Next we use price-setting equation to derive NKPC.

PMP and price-setting behavioral equation

Set base period on the intial period $t = 0$ for simplicity

$$\max_{P_{i0}} \pi_0^n = E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{P_{i0} Y_{it|0} - [TC_{it|0}^n(Y_{it|0})]\}$$
$$s.t.\ Y_{it|0} = (\frac{P_{i0}}{P_t})^{-\epsilon} Y_t \text{ (since } C_t = Y_t \text{ at equilbrium, the constraint is essentially demand curve)} \tag{344}$$

$|0$ means evaluated at prices from period 0
Y_{it} is total sales of firm i at period t
TC is a function of nominal total cost
$Q_{0,t}$ means discount rate from period 0 to period t
θ means the probability of price adjusting, therefore $1-\theta$ means the probablity to stay put. It's sticky price. [13]

Solving the PMP by putting demand curve into profit function, it would be

$$\pi_0^n = E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{P_{i0} (\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - [TC_{it|0}^n((\frac{P_{i0}}{P_t})^{-\epsilon} Y_t)]\} \tag{345}$$

then let the F.O.C. be 0 (fig 1,2)

$$\frac{\partial \pi_0^n}{\partial P_{i0}} = 0 \tag{346}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{(1-\theta)(\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - [\frac{\partial TC_{it|0}^n}{\partial P_{it|0}} \frac{\partial P_{it|0}}{\partial P_{i0}}] = 0 \tag{347}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - MC_{it|0}^n(-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon-1} Y_t \frac{1}{P_t}] = 0 \tag{348}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon) Y_t - MC_{it|0}^n(-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon-1} Y_t \frac{1}{P_t}] = 0 \tag{349}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon) Y_{it|0} + \frac{\epsilon}{P_t} MC_{it|0}^n (\frac{P_{i0}}{P_t})^{-1}] = 0 \tag{350}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} Y_{it|0} [(1-\epsilon) + \epsilon MC_{it|0}^n \cdot (P_{i0})^{-1}] = 0 \tag{351}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot (1-\epsilon) = -E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot \epsilon \cdot MC_{it|0}^n \cdot (P_{i0})^{-1} \tag{352}$$

$$\Rightarrow (1-\epsilon) E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} = -\epsilon (P_{i0})^{-1} E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot MC_{it|0}^n \tag{353}$$

restore Q_{0t} and $Y_{it|0}$ back to demand curve

$$\Rightarrow (\epsilon-1) E_0 \sum_{t=0}^{\infty} (1-\theta)^t \beta^t [(\frac{Y_0}{Y_t})^{\rho} \frac{P_0}{P_t}](\frac{P_{i0}}{P_t})^{-\epsilon} Y_t = (P_{i0})^{-1} E_0 \sum_{t=0}^{\infty} (1-\theta)^t \beta^t [(\frac{Y_0}{Y_t})^{\rho} \frac{P_0}{P_t}](\frac{P_{i0}}{P_t})^{-\epsilon} Y_t \cdot MC_{it|0}^n \tag{354}$$

[13] For more of price adjusting, see in Calvo, 1983 for uncertain version; Taylor, 1980 for certain version.

take P_{i0} and P_0 out since it can be treated as constants

$$\Rightarrow (\epsilon-1)\underline{P_{i0}^{-\epsilon}P_0Y_o^{\sigma}}E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\rho}=\epsilon\underline{P_{i0}^{-1}P_{i0}^{-\epsilon}P_0Y_0^{\rho}}E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n \tag{355}$$

eliminates the underlined common factors on RH and LH

$$\Rightarrow (\epsilon-1)E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\rho}=\epsilon E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n \tag{356}$$

$$\Rightarrow P_{i0}^*=\frac{\epsilon}{\epsilon-1}\frac{E_0\sum\limits_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n}{E_0\sum\limits_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}}\text{ (sticky price form price-setting equation)} \tag{357}$$

Also acquiring optimal price p^*, and $\frac{\epsilon}{\epsilon-1}$ is cost mutltiplier

$$\Rightarrow P_{i0}^*=P_0^*=\frac{\epsilon}{\epsilon-1}\frac{[(1-\epsilon)\beta]^0P_o^{\epsilon-1}Y_0^{1-\sigma}MC_{0|0}^n}{(1-\theta)^0\beta^0P_0^{\epsilon-1}Y_0^{1-\sigma}}=\frac{\epsilon}{\epsilon-1}MC_{0|0}^n\text{ (elastic form price-setting equation)} \tag{358}$$

For elastic form, the intuition is that firms are adjusting price at any time so there is no need to observe profit based on any specific period. So no summation symbol.

Question 2 (what if price is elastic?)**.**

elasticity means adjusting price at all time. it means price sticky price is $0 \Leftrightarrow 1-\theta=0$.

Note 12. *monopolistic competition and elasticity are not the same thing*

next try the period-t-based version in textbook (pic 3)

$$P_t^*=\frac{\epsilon}{\epsilon-1}\frac{E_t\sum\limits_{k=0}^{\infty}(1-\theta)^k\beta^{t+k}P_{t+k}^{\epsilon-1}Y_{t+k}^{1-\sigma}\cdot MC_{t+k|t}^n}{E_t\sum\limits_{k=0}^{\infty}(1-\theta)^k\beta^tP_{t+k}^{\epsilon-1}Y_{t+k}^{1-\sigma}} \tag{359}$$

(here the subscript i is gone because all firms are making unanimous decisions)

$$P_t^*=\frac{\epsilon}{\epsilon-1}MC_{t|t}^n \tag{360}$$

Note 13 (CRTS in DNK)**.** *if production function is CRTS, then all firms would also be making unanimous decisions regardless of time period position*

log-linearization of sticky price form price-setting behavioral equation
the method is still first order Taylor expansion, though bit trickier (pic 3, 4, 5)

dividing P_{t-1} on both side, and try seperately

$$\underbrace{\frac{P_t^*}{P_{t-1}} E_t \sum_{k=0}^{\infty} (1-\theta)^k \beta^t P_{t+k}^{\epsilon-1} Y_{t+k}^{1-\sigma}}_{\text{LHS}} = \underbrace{\frac{1}{P_{t-1}} \frac{\epsilon}{\epsilon-1} E_t \sum_{k=0}^{\infty} (1-\theta)^k \beta^{t+k} P_{t+k}^{\epsilon-1} Y_{t+k}^{1-\sigma} \cdot MC_{t+k|t}^n}_{\text{RHS}} \tag{361}$$

LHS mutil-variate first order Taylor expansion, the constant term is LHS at steady state where all P are equal

$$\begin{aligned}
\text{LHS's 1st order Taylor expansion} = & \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} \\
& + \frac{1}{p} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon} Y^{1-\sigma} (\frac{P_t^* - P}{P}) \\
& - \frac{P}{P^2} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{P_{t-1} - P}{P}) \\
& + (\epsilon-1) E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{P_{t+k} - P}{P}) \\
& + (1-\sigma) E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{Y_{t+k} - Y}{Y})
\end{aligned} \tag{362}$$

Like before, we use lowercase letters denotes log version

$$\Rightarrow \text{LHS} = P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 + \hat{p}_t^* - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k}] \tag{363}$$

For RHS,

$$\text{RHS} = \cdots = \frac{\epsilon}{\epsilon-1} \frac{1}{P} P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k} + \hat{mc}_{t+k|t}^n] \tag{364}$$

Note 14 (First order Taylor expansion and steady state)**.** *First term (constant term) of 1st Taylor expansion is steady state*

Question 3 (why do we do log-linearization)**.**

to simulate a curve with a line, so there would be a foundable or writable solution.

Now the two sides are log-linearized, let them be in the same equation

$$\underbrace{P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 + \hat{p}_t^* - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k}]}_{\text{log-linear LHS}} = \tag{365}$$

$$\underbrace{\frac{\epsilon}{\epsilon-1} \frac{1}{P} P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k} + \hat{mc}_{t+k|t}^n]}_{\text{log-linear RHS}} \tag{366}$$

Use sticky form price-setting equation at steady state (which because of steadiness looks like elastic form), we have

$$\frac{MC^n}{P} = \frac{\epsilon - 1}{\epsilon} \tag{367}$$

Since E_t is followed by an infinite series, it is meaningless. Eliminate it.

$$\Rightarrow \hat{p}_t^* \cancel{E_t} \sum_{k=0}^{\infty} [(1-\theta)\beta]^k = E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k \hat{mc}_{t+k|t}^n \tag{368}$$

$$\Rightarrow \frac{\hat{p}_t^*}{1-(1-\theta)\beta} = E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k MC_{t+k|t}^n \tag{369}$$

$$\Rightarrow \hat{p}_t^* = [1-(1-\theta)\beta] E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k \hat{mc}_{t+k|t}^n \text{ (log-linear sticky price setting equation)} \tag{370}$$

Therefore we have the log-linear optimal-price-setting equation for monopolistic competition and sticky price features firms.

8.2.2 Philips curve

First way to acquire Philips curve
next we use this log-linearized price-setting equation to acquire NK Philips curve
originally PC is about inflation and employment
step I, inflation (p7,8,9)

$$P_t = I(\frac{1}{I}\sum_{i=1}^{I} P_{it}^{\frac{1}{-\wedge_t}})^{-\wedge_t} \text{ (expression of total price)} \tag{371}$$

$$= [\int_0^{\theta} (P_{it}^*)^{1-\epsilon} di + \int_{\theta}^{1} P_{t-1}^{1-\epsilon} di]^{\frac{1}{1-\epsilon}} \text{ (θ portion of firms don't adjust price)} \tag{372}$$

$$= [\theta(P_{it}^*)^{1-\epsilon} + (1-\theta)P_{t-1}^{1-\epsilon}]^{\frac{1}{1-\epsilon}} \text{ (expression of total price with θ firms adjusting price)} \tag{373}$$

$$\Rightarrow P_t^{1-\epsilon} = \theta(P_{it}^*)^{1-\epsilon} + (1-\theta)P_{t-1}^{1-\epsilon} \tag{374}$$

$$\Rightarrow (\frac{P_t}{P_{t-1}})^{1-\epsilon} = \theta(\frac{P_t^*}{P_{t-1}})^{1-\epsilon} + (1-\theta) \tag{375}$$

$$\Rightarrow \Pi^{1-\epsilon} = \theta(\frac{P_t^*}{P_{t-1}})^{1-\epsilon} + (1-\theta) \text{ (the capital letter } \Pi \text{ is for total inflation } \frac{P_t}{P_{t-1}}) \tag{376}$$

Then log-linearization by bi-variate first order Taylor expansion

$$\Rightarrow \Pi^{1-\epsilon} + (1-\epsilon)\Pi^{1-\epsilon}(\frac{\Pi_t - \Pi}{\Pi}) = \{\theta(\frac{P}{P})^{1-\epsilon} + \theta(1-\epsilon)(\frac{P}{P})^{1-\epsilon}[\frac{1}{P}(P_t^* - P) - \frac{P}{P^2}\frac{(P_{t-1} - P)}{P}]\}^{1-\theta} \tag{377}$$

Since first term of tay exp is steady state, making $\Pi^{1-\epsilon} = 1$

$$\Rightarrow 1 + (1-\epsilon)\hat{\Pi}_t = \theta + \theta(1-\epsilon)(\hat{P}_t^* - \hat{P}_{t-1}) + (1-\theta) \tag{378}$$

$$\Rightarrow \hat{\Pi}_t = \theta(\hat{P}_t^* - \hat{P}_{t-1}) \tag{379}$$

$$\Rightarrow \frac{1}{\theta}\hat{\Pi}_t = \hat{P}_t^* - \hat{P}_{t-1} \tag{380}$$

Using price-setting equation, which is related to marginal cost, then production. (making nominal price combination of real price and $\hat{p}_{t-1}$)

$$\Rightarrow \frac{1}{\theta}\hat{\Pi}_t = [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{mc}^r_{t+k|t} + \hat{p}_{t+k} - \hat{p}_{t-1}) \tag{381}$$

$$= \underbrace{[1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k \hat{mc}^r_{t+k|t}}_{A} + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{p}_{t+k} - \hat{p}_{t-1}) \tag{382}$$

Let the untouched part be A for simplicity, open the sum so there will be expression of inflation in log-linearized form

$$= A + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{p}_{t+k} - \hat{p}_{t-1}) \tag{383}$$

$$\begin{aligned} &= A + [1-(1-\theta)\beta]E_t\{(\hat{p}_t - \hat{p}_{t-1}) + [(1-\theta)\beta](\hat{p}_{t+1} - \hat{p}_t + \hat{p}_t - \hat{p}_{t-1}) \\ &\quad + [(1-\theta)\beta]^2(\hat{p}_{t+2} - \hat{p}_{t+1} + \hat{p}_{t+1} - \hat{p}_t + \hat{p}_t - \hat{p}_{t-1}) + \dots\} \end{aligned} \tag{384}$$

$$= A + [1-(1-\theta)\beta]E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \tag{385}$$

$$\begin{aligned} &= A + E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \\ &\quad - (1-\theta)\beta E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \\ &\quad \text{(making elimination possible)} \end{aligned} \tag{386}$$

$$\begin{aligned} &= A + E_t\{\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2}\hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \\ &\quad - [(1-\theta)\beta]\{\hat{\pi}_t - [(1-\theta)\beta]^2 E_t(\hat{\pi}_{t+1} + \hat{\pi}_t) - \dots\} \end{aligned} \tag{387}$$

Make format simple by doing iteration form, the first term is when $k = 0$, but the second still starts from k=0 so there will be some parameters changed

$$= [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty} \hat{mc}^r_{t+k|t} + E_t \sum_{k=0}^{\infty} \hat{\pi}_{t+k+1} \tag{388}$$

$$= \underbrace{[1-(1-\theta)\beta]mc^r_{t+k|t} + \pi_t}_{k=0} + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^{k+1}\hat{mc}^r_{t+k+1|t} + E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^{k+1}\hat{\pi}_{t+k} \tag{389}$$

$$= [1-[(1-\theta)\beta]]\hat{mc}^r_{t|t} + \hat{\pi}_t + [(1-\theta)\beta]\{\qquad\} \tag{390}$$

$$= B \text{ (back to recursive form and let it be B)} \tag{391}$$

$$\Rightarrow \frac{1}{\theta}\hat{\pi}_t = B \tag{392}$$

$$\Rightarrow \hat{\pi}_t = \underbrace{\frac{\theta[1-(1-\theta)\beta]}{1-\theta}\hat{mc}^r_t}_{\text{effect of current marginal cost}} + \underbrace{\beta E_t\hat{\pi}_{t+1}}_{\text{effect of expectation of future inflation}} \quad \text{(New Keynesian Philips Curve marginal cost version)} \tag{393}$$

Question 4 (at which variable are we making derivative of Lagrangian function ?)**.**

Theoratically we can make derivative of any variable if possible but not proper.

NKPC has another version where marginal cost is substitued with real output, thus measuring the effect of output gap.

$$\begin{aligned} &\min \frac{W_t}{P_t} N_t \\ &s.t. Y_t = A_t N_t^{1-\alpha} \end{aligned} \tag{394}$$

Form a Lagrangian

$$\Rightarrow \mathscr{L} = \frac{W_t}{P_t} N_t + MC_t^r \cdot (Y_t - A_t N_t^{1-\alpha}) \text{ (shadow price can also be marginal cost)} \tag{395}$$

Take partial derivative of N_t here since labor is decided by market, the firm only can control demand of labor

$$\Rightarrow \frac{\partial \mathscr{L}}{\partial N_t} = 0 \tag{396}$$

$$\Rightarrow \frac{W_t}{P_t} N_t = MC_t^r \cdot MPN_t \text{ , where } MPN_t = \frac{\partial Y_t}{\partial N_t} = (1-\alpha) A_t N_t^{-\alpha} \tag{397}$$

$$\Rightarrow MC_t^r = \frac{\frac{W_t}{P_t}}{MPN_t} \tag{398}$$

$$\hat{mc}_t^\sigma = \hat{w}_t - \hat{p}_t - (a_t - \alpha \hat{n}_t) \text{ (log-linearized, and using labor supply equation)} \tag{399}$$

$$= \sigma \hat{c}_t + \varphi \hat{n}_t - (a_t - \alpha \hat{n}_t) \tag{400}$$

$$= \sigma \hat{y}_t - a_t - (\varphi + \alpha) \frac{\hat{y}_t - a_t}{1-\alpha} \tag{401}$$

$$= \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} \hat{y}_t - \frac{1+\varphi}{1-\alpha} a_t \tag{402}$$

To acquire natural ratio of output, we use the fact that elastic marginal cost is zero.

$$\Rightarrow 0 = \hat{mc}_t^r = \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} \hat{y}_t^f - \frac{1+\varphi}{1-\alpha} a_t \text{ (upscript } f \text{ stands for flexible meaning elastic)} \tag{403}$$

$$\Rightarrow \frac{1+\varphi}{1-\alpha} a_t = \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} \hat{y}_t^f \tag{404}$$

$$\Rightarrow \hat{mc}_t^r = \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} (\underbrace{\hat{y}_t - \hat{y}_t^f}_{\equiv \tilde{y}_t, \text{ output gap}}) \text{ (put it back, having the gap of output } \hat{y}_t - y_t^f = \tilde{t}) \tag{405}$$

If we put the last equation back to NKPC marginal cost version, we connect natural output to inflation which is NKPC real output version.

New Keynesian Philips Curve marginal cost version:

$$\hat{\pi}_t = \underbrace{\frac{\theta[1-(1-\theta)\beta]}{1-\theta} \frac{\sigma(1-\alpha) + (\varphi + \alpha)}{1-\alpha} (\hat{y}_t - \hat{y}_t^f)}_{\text{effect of current output gap}} + \underbrace{\beta E_t \hat{\pi}_{t+1}}_{\text{effect of expectation of future inflation}} \tag{406}$$

another way to acquire the NKPC is to concern monopolistic competition, then sticky price.

$$\max_{P_{it}} \pi^r = \frac{P_{it}}{P_t} Y_{it} - \frac{W_t}{P_t} N_{it} \tag{407}$$

$$\Rightarrow \frac{P_{it}}{P_t} = \frac{\epsilon}{\epsilon - 1} MC_t^r \tag{408}$$

Note 15 (indicator of monopolistic competition)**.**

$\frac{\epsilon}{\epsilon-1}$ means monopolistic competition
integral means perfect competitive labor supply for households
ϵ is subtitute elasticity

This is real price, where ϵ is subtitute elasticity. Here MC_i comes without subscript t is because of CRTS production function.

$$\Rightarrow \frac{P_{it}}{P_t} = \frac{\epsilon}{\epsilon - 1} \frac{W_t / P_t}{A_t} \tag{409}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma N_t^\varphi}{A_t} \tag{410}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t} (\int_0^1 N_{it} di)^\varphi \text{ (because of goods market p.c.)} \tag{411}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t} [\int_0^1 (\frac{Y_{it}}{A_t}) di]^\varphi \text{ (based on CRTS production function)} \tag{412}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t^{1+\varphi}} [\int_0^1 (\frac{P_{it}}{P_t})^{-\epsilon} Y_t di]^\varphi \text{ (using demand curve and move } A_t \text{ out)} \tag{413}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t^{1+\varphi}} (\frac{P_{it}^*}{P_t})^{-\epsilon\varphi} \text{ (star means symmetric and at equil)} \tag{414}$$

Since at equilibrium all prices are same

$$P_{it}^* = P_t^* \Rightarrow \frac{P_{it}^*}{P_t} = \frac{\epsilon}{\epsilon - 1} \frac{Y_t^{\sigma+\varphi}}{A_t^{1+\varphi}} (\frac{P_t^*}{P_t})^{-\epsilon\varphi} \tag{415}$$

Let's consider a situation that

$$\text{If } P_t^* = P_t \text{ (meaning when price elastic)} \tag{416}$$

$$1 = \frac{\epsilon}{\epsilon - 1} \tag{417}$$

$$\Rightarrow Y_t^f \text{ or } \hat{y}_t^f \text{ (to be log-linearized)} \tag{418}$$

Taking logs

$$(1 + \epsilon\varphi)(P_t^* - P_t) = \ln \frac{\epsilon}{\epsilon - 1} - (1 + \varphi) a_t + (\sigma + \varphi) y_t \tag{419}$$

$$= (\sigma + \phi) y_t - (1 + \phi) a_t + \ln(\frac{\epsilon}{\epsilon - 1}) \text{ (change order)} \tag{420}$$

Change order for furthur preparation

$$\Rightarrow P_t^* = P_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} y_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t + \frac{1}{1+\epsilon\varphi} \ln(\frac{\epsilon}{\epsilon-1}) \tag{421}$$

$$\Rightarrow P^* = P + \frac{\sigma+\varphi}{1+\epsilon\varphi} y_t - 0 + \frac{1}{1+\epsilon\varphi} \ln(\frac{\epsilon}{\epsilon-1}) \text{ (steady state by definition)} \tag{422}$$

Two equations above deducted, having the log-linearized result; elastic price means $p_t^* = \hat{p}_t$

$$\hat{p}_t^* = \hat{p}_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t \tag{423}$$

$$P_t^* = \hat{p}_t \text{ (elastic price gives } y \text{ a } f \text{ upscript)} \tag{424}$$

$$\Rightarrow 0 = \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t \tag{425}$$

$$\Rightarrow \hat{y}_t^f = \frac{1+\varphi}{\sigma+\varphi} a_t \tag{426}$$

here notice natural output is a function of technology

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t^f \tag{427}$$

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \gamma(\hat{y}_t - \hat{y}_t^f) \text{ (γ denotes the bundle of coefficients)} \tag{428}$$

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \gamma\tilde{y}_t \text{ (acceptable price-setting euqation with only assumption of m.c.)} \tag{429}$$

14

Note 16 (DNK model is perfect information rational expectation)**.** *Here E_t is not $\bar{E}_t$ because of perfect information rational expectation*

Second way to acquire NKPC

We need to know two methods of price-setting:

- Price adjusting in certainty, Taylor, 1978[15]
- price adjusting in randomness, Calvo, 1983[16]

Note 17 (Price adjusting in certainty vs in randomness)**.**

Taylor

a, b and c denotes three firms in a monopolistic competitive market. they adjust price by turns and one person per period.

Q1: how does one choose price

$P_A^{\triangle} = \frac{}{3}$, the upscript $\triangle$ stands for staying put in the next three periods.

[14] See Mankiw and Reis 2002 for more details of acceptable price-setting equation

[15] Taylor, John B. "Staggered Wage Setting in a Macro Model." The American Economic Review 69, no. 2 (1979): 108–13.

[16] Calvo, Guillermo A., 1983. "Staggered prices in a utility-maximizing framework," Journal of Monetary Economics, Elsevier, vol. 12(3), pages 383-398, September.

Q2: what is the total price
P_t =
Calvo
each firm adjust price at probablity θ^t at period t.

Now sticky price price-adjusting equation

$$\hat{p}_t^{\triangle} = \underbrace{(1-\theta\beta)\sum_{k=0}^{\infty}(\theta\beta)^k E_t P_{t+k}^*}_{\text{weighted average}} \tag{430}$$

$$\Rightarrow \hat{p}_t^{\triangle} = (1-\theta\beta)\hat{p}_t^* + \theta\beta E_t \hat{p}_{t+1}^{\triangle} \text{ (sticky PAE)} \tag{431}$$

The θ is the opposite of the one in previous PAE, here rather is to stay put. price adjusting based on acceptable price setting equation in future periods, β mean subjective discount.

With the expression of total price being

$$\Rightarrow \hat{p}_t = (1-\theta)\sum_{k=0}^{\infty}\theta^k \hat{p}_{t-k}^{\triangle} \tag{432}$$

$$\Rightarrow \hat{p}_t = (1-\theta)\hat{p}_t^{\triangle} + \theta\hat{p}_{t-1} \text{ (expression of total price based on APSE and PAE)} \tag{433}$$

Based on Acceptable Price-setting Equation (from monopolistic competition) and Total Price (from sticky price), one can quickly conduct NKPC.

Note 18 (prices in DNK)**.**

starred price means acceptable price, triangled price means adjusting price

From the first equation $\hat{p}_t = (1-\theta)\sum_{k=0}^{\infty}\theta^k \hat{p}_{t-k}^{\triangle}$ we have

$$\hat{p}_t = (1-\theta)P_t^{\triangle} + \theta P_{t-1} \tag{434}$$

$$\Rightarrow (1-\theta)P_t^{\triangle} = \hat{p}_t - \theta\hat{p}_{t-1} \tag{435}$$

$$\Rightarrow (1-\theta)E_t P_{t+1}^{\triangle} = E_t\hat{p}_{t+1} - \theta\hat{p}_t \tag{436}$$

$$\Rightarrow (1-\theta)\beta E_t \hat{P}_{t+1}^{\triangle} = \beta E_t \hat{p}_{t+1} - \theta\beta\hat{p}_t \tag{437}$$

From the second equation $\hat{p}_t = (1-\theta)\hat{p}_t^{\triangle} + \theta\hat{p}_{t-1}$, first use sticky price price-adjusting equation $\hat{p}_t^{\triangle} = (1-\theta\beta)\hat{p}_t^* + \theta\beta E_t\hat{p}_{t+1}^{\triangle}$ to substitute

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)\hat{p}_t^{\triangle} = (1-\theta)(1-\theta\beta)\hat{p}_t^* + (1-\theta)\theta\beta E_t P_{t+1}^{\triangle} \tag{438}$$

then use APSE $\hat{p}_t^* = \hat{p}_t + \gamma\tilde{y}_t$ to substitute

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)(1-\theta\beta)(\hat{p}_t + \gamma\tilde{y}_t) + (1-\theta)\theta\beta E_t P_{t+1}^{\triangle} \tag{439}$$

use the conclusion $(1-\theta)E_t P_{t+1}^{\triangle} = E_t\hat{p}_{t+1} - \theta\hat{p}_t$ above

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)(1-\theta\beta)(\hat{p}_t + \gamma\tilde{y}_t) + \theta\beta E_t\hat{p}_{t+1} - \theta^2\beta\hat{p}_t \tag{440}$$

$$\Rightarrow [1+\theta^2\beta - (1-\theta)(1-\theta\beta)]\hat{p}_t - \theta P_{t-1} = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta E_t \text{ (break APSE)} \tag{441}$$

$$\Rightarrow [1+\theta^2\beta - (1-\theta)(1-\theta\beta)]\hat{p}_t - \theta P_{t-1} - \theta\beta\hat{p}_t = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta(E_t\hat{p}_{t+1} - \hat{p}_t) \tag{442}$$

$$\Rightarrow \theta(\hat{p}_t - \hat{p}_{t-1}) = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta(E_t\hat{P}_{t+1} - \hat{p}_t) \text{ (match terms so there is inflation)} \tag{443}$$

$$\Rightarrow \theta\hat{\pi}_t = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta E_t\hat{\pi}_{t+1} \text{ (here comes inflation)} \tag{444}$$

$$\Rightarrow \hat{\pi}_t = k\hat{y}_t + \beta E_t\hat{\pi}_{t+1} \text{ (NKPC)} \tag{445}$$

8.2.3 Dynamic IS curve

Recalling that household

$$\hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \tag{446}$$

$$\Rightarrow \hat{y}_t^f = E_t\hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - 0) \text{ (inflation elastic form is assumed as 0)} \tag{447}$$

$$\Rightarrow \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \text{ (dynamic IS curve, DIS)} \tag{448}$$

8.3 Equilibrium of model

Now NKPC and DIS each represents supply and demand. The two equations have three unkown variables $\hat{\pi}_t, \tilde{y}_t, \hat{i}_t$, applying the model does not have a solution. It is so that we need to add another variable into the model. There are two ways making DNK model solvable, if not monetary supply rule then Taylor rule.

8.3.1 DNK with money supply rule

$$\text{DNK} \begin{cases} \hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)} \\ \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \text{ (DIS)} \\ \hat{m}_t = \hat{p}_t + \hat{y}_t - \eta\hat{i}_t \text{ (Money supply rule, used to be ad hoc in RBC)} \end{cases} \tag{449}$$

Seems now three equations have even more unkown variables $\hat{\pi}_t, \tilde{y}_t, \hat{i}_t, \hat{p}_t, m_t$.

But hat $\pi_t, \hat{p}_t$ are the same thing because

$$\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1} \tag{450}$$

$$\Rightarrow \hat{p}_t = \hat{\pi}_t + \hat{p}_{t-1} \tag{451}$$

$$\Rightarrow \hat{m}_t = \hat{\epsilon}_t \tag{452}$$

this transfer works either way.

Let the m_t money supply be exogenous for simplicity.

Note 19 (Exogenous variables in DNK model)**.** *Only technology and money supply are considered real exogenous variables here.*

So we are left with three equations and three unkowns, the model is now solutionable. The final result will be a function of m_t.

8.3.2 DNK with interest rate rule

$$\text{DNK}\begin{cases}\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)}\\ \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \text{ (DIS)}\\ \hat{i}_t = \phi_\pi\hat{\pi}_t + \phi_y\hat{y}_t + \nu_t \text{ (Interest rate rule)}\end{cases} \tag{453}$$

$$\text{where } \nu_t = \rho_v\nu_{t-1} + \epsilon_t^\nu \text{ is the exogenous shock of interest rate, with the feature of AR1} \tag{454}$$

Note 20 (Interest rate rule)**.**

$$\hat{i}_t = \phi_\pi \cdot (\hat{\pi}_t - \hat{\pi}^*) + \phi_y \cdot (\hat{y}_t - y_t^f) + \nu_t \tag{455}$$

If the target parameters $\hat{\pi}^*$ and y_t^f are given, it is Taylor Rule.
Operation parameters ϕ_π and ϕ_y are decided based on past experience.

Acquiring equilibrium and prove it is the only equilibrium
Rewrite the DIS

$$\Rightarrow \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}[(\phi_\pi\hat{\pi}_t + \phi_y\hat{y}_t + \nu_t) - E_t\hat{\pi}_{t+1}] \text{ (apply interest rate rule to the second equation)} \tag{456}$$

$$= E_t\tilde{y}_{t+1} - \frac{1}{\sigma}\{[\phi_\pi \cdot (\beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t) + \phi_y\hat{y}_t + \nu_t] - E_t\hat{\pi}_{t+1}\} \text{ (put the first in)} \tag{457}$$

$$= E_t\tilde{y}_{t+1} + \frac{1-\beta\phi_\pi}{\sigma}E_t\pi_{t+1} - \frac{\phi_\pi k + \phi_y}{\sigma}\tilde{y}_t - \frac{\nu_t}{\sigma} \tag{458}$$

$$= \frac{1}{\sigma + \phi_\pi k + \phi_y}[\sigma E_t\tilde{y}_{t+1} + (1-\beta\phi_\pi)E_t\hat{\pi}_{t+1} - \nu_t] \text{ (the relationship of } y_t \text{ and } y_{t+1} \text{ and } \pi_{t+1}) \tag{459}$$

Similarly we can change the NKPC

$$\hat{\pi}_t = \frac{1}{\sigma + \phi_\pi k + \phi_y}\{k\sigma E_t\tilde{y}_{t+1} + [k + \beta(\sigma + \phi_y)]E_t\hat{\pi}_{t+1} - \nu_t\} \tag{460}$$

Therefore three equations are two now.

Iteration in vector form

$$\begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\end{bmatrix} = \frac{1}{\sigma+\phi_\pi k+\phi_y}\begin{bmatrix}\sigma & 1-\beta\phi_\pi\\ k\sigma & k+\beta(\sigma+\phi_y)\end{bmatrix}\begin{bmatrix}E_t\hat{y}_{t+1}\\ E_t\hat{\pi}_{t+1}\end{bmatrix} - \frac{1}{\sigma+\phi_\pi k+\phi_y}\begin{bmatrix}1\\ k\end{bmatrix}\nu_t \tag{461}$$

$$\Rightarrow \begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\end{bmatrix} = A\begin{bmatrix}C_{t-1}\\ K_{t-1}\end{bmatrix} - B\nu_t \text{ (at RHS one period ahead meaning the whole equation is foresight-seeing)} \tag{462}$$

$$\Rightarrow \lambda^2 - \frac{\sigma+k+\beta(\sigma+\phi_y)}{\sigma+\phi_\pi k+\phi_y}\lambda + \frac{\sigma\beta}{\sigma+\phi_\pi k+\phi_y} = 0 \tag{463}$$

Note 21 (Technique in Ramsey and DNK)**.**

From the perspective of techniques in persuiing solution, it is greatly similar to the way we did with Ramsey model, where the model is one period backward at RHS.

Note 22 (HNKPC)**.** *Hybrid New Keynesian Philips Curve*

Since NKPC lacks persistence in practical use (shocks are too slow to effects), we might as well add a 1-period-lagged inflation to the RHS as adpative expectation. This is somehow a nominal shock since it gets into the DNK model through m_t.

$$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)} \tag{464}$$

$$\hat{\pi}_t = \hat{\pi}_{t-1} + \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (HNKPC)} \tag{465}$$

a_t is described as real shocks of DNK model. But it shows up too fast.
Later the concept of sticky information is added to the model to solve the problem.

Note 23 (BK conditions)**.**

First brought up by Blanchard and Kahn, 1980

[17]

Note 24.

Lasalle, 1986

$$x^2 + bx + c = 0 \tag{466}$$

$$x_1, x_2 < 1 \iff \begin{cases}|c| < 1\\ |b| < 1 + c\end{cases} \tag{467}$$

[17] Also see Nakajima 2007 for similar techniques.

If the equilibrium exists and it is the only one, then

$$\begin{cases} |\frac{\sigma\beta}{\sigma+\phi_\pi k+\phi_y}| < 1 \Rightarrow \sigma(\beta-1) < 0 < \phi_\pi k + \phi_y \\ |\frac{\sigma+k+\beta(\sigma+\phi_y)}{\sigma+\phi_\pi k+\phi_y}| < 1 + \frac{\sigma\beta}{\sigma+\phi_\pi k+\phi_y} \Rightarrow k(\phi_\pi - 1) + (1-\beta)\phi_y > 0 \end{cases} \tag{468}$$

Method of Undetermined Coefficients:
Let's assume

$$\begin{cases} \tilde{y}_t = \psi_{y\nu}\nu_t \\ \hat{\pi}_t = \psi_{\pi\nu}\nu_t \end{cases} \tag{469}$$

recall that

$$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\hat{y}_t \text{ (expected inflation)} \tag{470}$$

$$\psi_{\pi\nu}\nu_t = \beta E_t(\psi_{\pi\nu}\nu_{t+1}) + k\hat{y}_t \text{ (iterating)} \tag{471}$$
$$= \beta E_t[\psi_{\pi\nu}(\rho_\nu\nu_t + \epsilon^\nu_{t+1})] + k\hat{y}_t \text{ (expand } \nu_{t+1} \text{ since it's AR1 shock of } i_t) \tag{472}$$
$$= \beta\psi_{\pi\nu}\rho_\nu\nu_t + k\psi_{y\nu}\nu_t \text{ (preassuming expectation of next period noise is } 0) \tag{473}$$
$$\Rightarrow (1-\beta\rho_\nu)\psi_{\pi\nu} = k\psi_{y\nu} \tag{474}$$
$$\Rightarrow k\psi_{y\nu} = \frac{1-\beta\rho_\nu}{k}\psi_{\pi\nu} \text{ (relationship of two undetermined coefficients)} \tag{475}$$

Using DIS

$$\tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}[\phi_\pi\hat{\pi}_t + \phi_y\hat{y}_t + \nu_t) - E_t\hat{\pi}_{t+1}] \tag{476}$$

using interest rate rule

$$\Rightarrow \psi_{y\nu}\nu_t = \psi_{y\nu}E_t\nu_{t+1} - \frac{1}{\sigma}(\phi_\pi\psi_{\pi\nu}\nu_t + \phi_y\psi_{y\nu}\nu_t + \nu_t - \psi_{\pi\nu}E_t\nu_{t+1}) \tag{477}$$
$$= \psi_{y\nu}\rho_\nu\nu_t - \frac{1}{\sigma}(\phi_\pi\psi_{\pi\nu}\nu_t + \phi_y\psi_{y\nu}\nu_t + \nu_t - \psi_{\pi\nu}\rho_\nu\nu_t) \tag{478}$$
$$= \frac{\sigma\rho_\nu - \phi_y}{\sigma}\phi_{y\nu}\nu_t - \frac{\phi_\pi - \rho_\nu}{\sigma}\psi_{\pi\nu}\nu_t - \frac{1}{\sigma}\nu_t \tag{479}$$
$$\Rightarrow \sigma\psi_{y\nu} = (\sigma\rho_\nu - \phi_y)\psi_{y\nu} - (\phi_\pi - \rho_\nu)\psi_{\pi\nu} - 1 \text{ (elimination of } \nu_t) \tag{480}$$
$$\Rightarrow (\sigma - (\sigma\rho_\nu - \phi_y))\psi_{y\nu} + (\phi_\pi - \rho_\nu)\psi_{\pi\nu} = -1 \tag{481}$$
$$\Rightarrow \psi_{\pi\nu} = \frac{-k}{\cdots} = -k\wedge_\nu \tag{482}$$
$$\Rightarrow \psi_{y\nu} = -(1-\beta\rho_\nu)\wedge_\nu \tag{483}$$

8.4 Analyze exogenous shocks

Now policy change will reflect in the parameter ϵ, making policy analysis possible.
we use ν to denote the exogenous shock
last section we have expression of ν
$\Rightarrow$ to analyze the shock would mean fully using it.

Note 25 (what is a shock?)**.**

In a model with uncertainty, variable has the feature of causing change in variation by change itself can cause a shock.

But it was later broadened as change of exogenous variable in a model without uncertainty.

recall output gap at equilibrium

$$\begin{aligned}
\hat{y}_t - \hat{y}_t^f = \hat{y}_t^* &= \psi_{y\nu}\nu_t && (484)\\
&= -(1-\beta\rho_\nu) \wedge_\nu \nu_t && (485)\\
&= -(1-\beta\rho_\nu) \wedge_\nu (\rho_\nu \nu_{t-1} + \epsilon_t^\nu) && (486)\\
&= -(1-\beta\rho_\nu) \wedge_\nu \rho_\nu\nu_{t-1} - (1-\beta\rho_\nu) \wedge_\nu \epsilon_t^\nu && (487)\\
&= \rho_\nu \tilde{y}_{t-1} - (1-\beta\rho_\nu) \wedge_\nu \epsilon_t^\nu \ (\tilde{y}_{t-1} \text{ backward one period}) && (488)
\end{aligned}$$

Note 26. *we can only do analysis at equil*

similiar with inflation at equilibrium

$$\begin{aligned}
\hat{\pi}_t^* &= \psi_{\pi\nu}\nu_t && (489)\\
&= -k \wedge_\nu \nu_t && (490)\\
&= -k \wedge_\nu (\rho_\nu \nu_{t-1} + \epsilon_t^\nu) && (491)\\
&= \rho_\nu \pi_{t-1} - k \wedge_\nu \epsilon_t^\nu && (492)
\end{aligned}$$

interest rate at equilibrium

$$\begin{aligned}
\hat{r}_t^* &= \hat{i}_t - E_t\hat{\pi}_{t+1} \text{ (Real interest rate at equilibrium)} && (493)\\
&= \sigma(E_t\tilde{y}_{t+1} - \hat{y}_t) \leftarrow \text{DIS} && (494)\\
&= \sigma(1-\beta\rho_\nu)(1-\rho_\nu) \wedge_\nu \nu_t && (495)\\
\Rightarrow \hat{i}_t^* &= \hat{r}_t + E_t\hat{\pi}_{t+1} \text{Nominal interest rate at equilibrium} && (496)\\
&= [\sigma(1-\beta\rho_\nu)(1-\rho_\nu) - k\rho_\nu] \wedge_\nu \nu_t && (497)
\end{aligned}$$

moreever we can do analysis with money supply using quantity equation of money

$$\begin{aligned}
\hat{m}_t^* &= \hat{p}_t^* + \hat{y}_t^* - \eta\hat{i}_t^* \text{ (LHS is solvable since all variables at RHS has been solved)} && (498)\\
&= \text{ (assuming } \hat{y}_t^{f*} = 0) && (499)\\
&= && (500)\\
&= \text{ use solutions from above} && (501)\\
&\Rightarrow && (502)\\
&= && (503)\\
\frac{d\hat{m}_t^*}{d\nu_t} &= && (504)\\
&= \text{showing the result is decided by mutiple joint forces} && (505)
\end{aligned}$$

Note 27 (quantity equation of money)**.** $MV = Py$

While this theory was originally formulated by Polish mathematician Nicolaus Copernicus in 1517, it was popularized later by economists Milton Friedman and Anna Schwartz after the publication of their book, "A Monetary History of the United States, 1867-1960," in 1963.

8.5 Natural rates, output gaps, and model limits

Planned section. Distinguish the natural real interest rate, flexible-price output, and efficient output. Check which shocks and distortions enter each object before interpreting policy responses or welfare losses.

9 Sticky Wage Model

Before good market is price-sticky and monopolistic competitive, now the setting goes to labor market.

$$Y_{it} = (\frac{P_{it}}{P_t})^{-\epsilon_P} Y_t \text{ (good demand curve, } \epsilon \text{ is substitute elasticity)} \tag{506}$$

$$P_t = (\int_0^1 P_{it}^{1-\epsilon} di)^{\frac{1}{1-\epsilon_p}} \text{ (total price)} \tag{507}$$

$$W_t = W_{it} = \int_0^1 W_{it} di \tag{508}$$

$$P_{it}^* = c + \alpha \tilde{y}_t \text{ (desired price)} \tag{509}$$

labor market m.c. (as in classical model) and sticky wage (partial rigidity, in contrary to total rigidity where all wages are rigid).

1) labor demand curve and total wage price level

$$\begin{aligned} &\min \int_0^1 W_{jt} N_{ijt} dj \\ &s.t. A_t [(\int_0^1 N_{ijt}^{\frac{\epsilon_W - 1}{\epsilon_W}} dj)^{\frac{\epsilon_W}{\epsilon_W - 1}}]^{1-\alpha} \geqslant A_t N_{it}^{1-\alpha} \end{aligned} \tag{510}$$

where i indexing firm, j indexing labor, t is time

Form a Lagrangian

$$\mathscr{L} = \int_0^1 W_{jt} N_{ijt} dj + W_t \{A_t [(\int_0^1 N_{ijt}^{\frac{\epsilon_w - 1}{\epsilon_w}} dj)^{\frac{\epsilon_W}{\epsilon_W - 1}}]^{1-\alpha} - A_t N_{it}^{1-\alpha}\} \tag{511}$$

where Lagrangian multiplier is total wage level

Solving the optimization problem results

$$\Rightarrow \begin{cases} N_{ijt} = (\frac{W_{jt}}{W_t})^{-\epsilon_W} N_{it} \text{ (labor supply equation)} \\ W_t = (\int_0^1 W_{jt}^{1-\epsilon_W} dj)^{\frac{1}{1-\epsilon_W}} \text{ (expression of total wage level)} \end{cases} \tag{512}$$

2) UMP of household

in utility function N is labor demand instead of labor supply as before. this is because of m.c. labor market.

$$\begin{aligned} u &= u(C_t, N_t) \\ &= u(C_t, \{N_{jt}\}) \\ &= \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \int_0^1 \frac{N_j^{1+\varphi}}{1+\varphi} dj \text{ (CES style)} \end{aligned} \tag{513}$$

the UMP can be rewritten as

$$\begin{aligned} \max_{\{C_t, W_t^*\}} \sum_{t-0}^{\infty} \beta^t u(C_t, \int_0^1 N_{jt} dj) \iff \max \sum_{t=0}^{\infty} \beta^t u(C_t, \int_0^1 (\frac{W_t^*}{W_t})^{-\epsilon_W} N_{it} dj) \\ s.t. P_t C_t + Q_t B_t \leqslant \int_0^1 W_t^* (\frac{W_t^*}{W_t})^{-\epsilon_W} N_{it} dj + B_{t-1} \end{aligned} \tag{514}$$

here W_t^* instead of original because all wages can be adjusted, eventually they will be the same

a simplified version would be

$$\begin{aligned} &\max \sum_{t=0}^{\infty} \beta^t u(C_t, (\frac{W_t^*}{W_t})^{-\epsilon_W} N_{it}) \\ &s.t. P_t C_t + Q_t B_t \leqslant W_t^* (\frac{W_t^*}{W_t})^{-\epsilon_W} + B_{t-1} - T_t \end{aligned} \tag{515}$$

$$\begin{aligned} &\max_{W_0^*} \sum_{t=0}^{\infty} \beta^t \theta_W^t u(C_{t|0}, N_{t|0}) \\ &s.t. \begin{cases} C_{t|0} = \frac{1}{P_t}[W_0^* (\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it} + B_{t-1|0} - T_{t|0} - Q_{t-1|t} B_{t|0}] \\ N_{t|0} = (\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it} \text{ (with labor demand curve, left with only } W_0^*) \end{cases} \end{aligned} \tag{516}$$

sticky wage, where $\theta \equiv Prob(\text{stay put})$. θ is before utility function because the whole equation is based on period 0, entailing a prob for the whole equation to stay put.

put two constraints in the max part, now the optimization problem is a unconstrainted one

$$\max_{W_0^*} \sum_{t=0}^{\infty} (\beta\theta_W)^t u(\frac{1}{P_t}[W_0^* (\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it} + B_{t-1|0} - T_{t|0} - Q_{t-1|t} B_{t|0}], (\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it}) \tag{517}$$

$$FOC \Rightarrow E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t [u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} (1-\epsilon_W)(\frac{W_0^*}{W_t})^{-\epsilon_W} N_{it} - u_N(C_{t|0}, N_{t|0}) \epsilon_W (\frac{W_0^*}{W_t})^{-(\epsilon_W)^{-1}} \frac{1}{W_t} N_{it})] \tag{518}$$

$$= E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t [u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} (1-\epsilon_W)(\frac{W_0^*}{W_t})^{-\epsilon_W} W_t^{\epsilon_W} N_{it} - u_N(C_{t|0}, N_{t|0}) \epsilon_W (\frac{W_0^*}{W_t})^{-(\epsilon_W)} W_t^{\epsilon_W} N_{it} \frac{1}{W_0^*})] \tag{519}$$

move some exogenous variables out since they basically can be treated as constants

$$\Rightarrow (1-\epsilon_W)(W_0^*)^{-\epsilon_W} \ldots \tag{520}$$

$$= \epsilon_W (W_0^*)^{-\epsilon_W^{-1}} \tag{521}$$

$$\Rightarrow W_0^* = \frac{\epsilon_W}{1-\epsilon_W} \frac{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_N(C_{t|0}, N_{t|0}) W_t^{\epsilon_W} N_{it}}{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} W_t^{\epsilon_W} N_{it}} \tag{522}$$

use marginal substitute ratio, where $MRS_{t|0} \equiv -\frac{U_N(C_{t|0})}{U_C(C_{t|0})}$, the equation is essentially an inflation equation

$$W_0^* = \frac{\epsilon_W}{\epsilon_W - 1} \frac{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_N(C_{t|0}, N_{t|0}) MRS_{t|0} W_t^{\epsilon_W} N_{it}}{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} W_t^{\epsilon_W} N_{it}} \tag{523}$$

$$\Rightarrow \frac{W_0^*}{P_0} = \frac{\epsilon_W}{\epsilon_W - 1} MRS_{t|0} \text{ (if elastic wage and m.c., there will be no equation above)} \tag{524}$$

now based on the perspective at period t

$$W_t^* = \frac{\epsilon_W}{\epsilon_W - 1} \frac{E_t \sum_{k=0}^{\infty} (\beta\theta_W)^k u_C(C_{t+k|t}, N_{t+k|t}) MRS_{t+k|t} W_{t+k|t}^{\epsilon_W}}{E_t \sum_{k=0}^{\infty} (\beta\theta_W)^k u_C(C_{t+k|t}, N_{t+k|t}) \frac{1}{P_{t+k}} W_{t+k|t}^{\epsilon_W}} \text{ (at t period instead 0 period)} \tag{525}$$

$$\Rightarrow \frac{W_t^*}{P_t} = \frac{\epsilon_W}{\epsilon_W - 1} MRS \text{ (if elasticity wage and monopolistic competitive at t period)} \tag{526}$$

Note 28 (M.C. firm choice variable for optimization problem)**.** *MC firm can choose price max or quantity max*

10 Labor Markets and Unemployment

Planned chapter.

10.1 Employment, unemployment, and participation

Define the margins hidden by a single aggregate labor-supply variable. Distinguish changes in hours from movements into and out of employment.

10.2 Search, matching, and wage determination

Introduce a basic matching framework to explain vacancies and unemployment. State how wage setting and hiring incentives determine job creation.

10.3 Adjustment and persistence

Compare search frictions with nominal wage rigidity. Examine which mechanisms can make employment recover slowly and connect a recession to longer-run output.

Part IV
Open Economy

An open economy can trade goods and claims on future income. This changes the connection between domestic saving and investment, introduces relative prices between countries, and allows shocks to cross borders. We extend the earlier models before comparing policy choices in Part V.

11 International Accounts and Intertemporal Trade

Planned chapter.

11.1 Trade balances, current accounts, and external wealth

Derive the links between national saving, investment, and the current account. Distinguish transaction flows from valuation changes in the international investment position.

11.2 Consumption smoothing and international borrowing

Start with a small open endowment economy. Derive its intertemporal budget constraint and explain how expected income affects the current account.

11.3 Investment and external adjustment

Add capital accumulation and examine how borrowing finances investment. State borrowing and solvency conditions before discussing external adjustment.

12 Trade and Relative Prices

Planned chapter.

12.1 Comparative advantage, trade gains, and trade costs

Use a simple trade model to explain specialization and gains from exchange. Introduce tariffs and other trade costs to connect trade policy to allocation.

12.2 Terms of trade and the real exchange rate

Define both relative prices with explicit conventions. Trace how their changes affect purchasing power, production, and external balances.

12.3 Tradable goods, nontradable goods, and adjustment

Introduce the distinction between tradable and nontradable production. Explain how expenditure and resources move between sectors during adjustment.

13 Open Economy Fluctuations and Exchange Rates

Planned chapter.

13.1 The small open economy RBC model

Extend the real benchmark to international borrowing and investment. Compare productivity, world-interest-rate, and terms-of-trade shocks.

13.2 Capital flows, interest parity, and risk

Derive the relevant asset-pricing conditions and distinguish covered parity from an uncovered expected-return condition. Introduce risk premia and foreign-currency balance-sheet exposure.

13.3 Nominal rigidities and exchange-rate regimes

Introduce an open economy New Keynesian benchmark. Compare the adjustment mechanisms under fixed and flexible exchange rates, and state the constraints created by capital mobility.

Part V
Money, Finance, and Macroeconomic Policy

The earlier parts establish the real and nominal benchmarks. We now add financial constraints and differences between households, then compare the transmission and welfare effects of monetary and fiscal policy. Open economy mechanisms remain part of that comparison.

14 Money, Banking, and Central Bank Operations

Planned chapter.

14.1 Money, payments, and financial claims

Distinguish money from other assets and identify the services that give it a role in household and firm decisions.

14.2 Banks and balance sheets

Set out the balance sheets of households, commercial banks, and the central bank. Explain lending, deposits, reserves, and the limits to credit creation.

14.3 Implementing monetary policy

Connect the model's policy rate to the central bank's operating framework. Distinguish reserve provision, asset purchases, and changes in borrowing costs.

15 Financial Frictions and Aggregate Adjustment

Planned chapter.

15.1 Borrowing limits and collateral

Derive a simple borrowing constraint and show how it changes optimal choices. Connect collateral values to spending and investment.

15.2 Intermediation and financial amplification

Introduce a constraint on intermediary balance sheets. Explain how losses, leverage, and credit spreads can amplify a real or financial disturbance.

15.3 Financial distress and policy intervention

Distinguish liquidity problems from insolvency. State the mechanisms and trade-offs behind liquidity support, restructuring, and macroprudential policy.

16 Household Heterogeneity and Policy Transmission

Planned chapter.

16.1 Income risk, insurance, and precautionary saving

Introduce uninsurable income risk and borrowing limits. Explain how the distribution of liquid wealth affects saving and consumption.

16.2 Marginal propensities to consume and redistribution

Compare households' spending responses to income and wealth changes. Trace how taxes, transfers, and interest payments redistribute resources.

16.3 From representative to heterogeneous households

Use a two-agent benchmark before motivating a richer distribution of households. Identify which policy questions require the additional heterogeneity.

17 Monetary Policy

This chapter extends the interest-rate rules used to close the New Keynesian model into a systematic policy comparison. The existing welfare derivation is retained below; the preceding sections are outlines.

17.1 Transmission and policy shocks

Planned section. Separate a systematic policy response from an unexpected policy disturbance. Compare intertemporal, credit, redistribution, and exchange-rate channels.

17.2 Rules, commitment, and discretion

Planned section. Compare equilibrium determination with welfare-based policy design. Explain how credibility and promises about future policy affect current choices.

17.3 The interest-rate lower bound and balance-sheet policies

Planned section. State when the policy-rate constraint binds and which transmission channels remain. Examine forward guidance and asset purchases under explicit assumptions.

17.4 Optimal monetary policy

A significant paper Mankiw and Reis 2002 used a new keynesian macroeconomic model to acquire the optimal monetary policy. [18]

Note 29 (why is there welfare loss?)**.**

monopoly and sticky price cause deviation from Pareto optimality.

Note 30 (Optimal monetary policy)**.**

No concept of optimality can be found in policy analysis in the perspective of household or firm. But the government can use welfare loss fucntion in its optimality to maximize social welfare. In the process of achieving maximized social welfare, the monetary policy is the so-called optimal monetary policy.

To acquire the optimal monetary policy, the welfare loss function must be previously introduced.
To acquire WLF, we use second order taylor expansion
first order Taylor expansion

$$\frac{X_t - X}{X} = \ln X_t - \ln X = x_t - x = \hat{x}_t \tag{527}$$

[18] N. Gregory Mankiw, Ricardo Reis, Sticky Information versus Sticky Prices: A Proposal to Replace the New Keynesian Phillips Curve, The Quarterly Journal of Economics, Volume 117, Issue 4, November 2002, Pages 1295–1328, https://doi.org/10.1162/003355302320935034

second order Taylor expansion

$$\begin{aligned}\frac{X_t - X}{X} &= \frac{1}{X}e^{x_t} - 1 \\ &= (\frac{1}{X}e^{x} - 1) + \frac{1}{X}e^{x}(x_t - x) + \frac{1}{2}\frac{1}{X}e^{x}(x_t - x)^2 \\ &= \hat{x}_t + \frac{1}{2}\hat{x}_t^2\end{aligned} \tag{528}$$

Similarly

$$\Rightarrow \begin{cases} \frac{C_t - C}{C} \approx \hat{c}_t + \frac{1}{2}\hat{c}_t^2 \\ \frac{N_t - N}{N} \approx \hat{n}_t + \frac{1}{2}\hat{n}_t^2 \end{cases} \tag{529}$$

Bi-variate 2o taylor expansion for a utility function where people gain utilities from consuming goods and enjoying leisure.

$$u(C_t, N_t) \approx u(C, N) + u_C^C(\frac{C_t - C}{C}) + u_N^N(\frac{N_t - N}{N}) + \frac{1}{2}u_{CC}^{C^2}\frac{(C_t - C)^2}{C} + \frac{1}{2}u_{NN}^{N^2}\frac{(N_t - N)^2}{N} \tag{530}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + u_N N(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) + \frac{1}{2}u_{CC}C^2(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + \frac{1}{2}u_{NN}N^2(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) \tag{531}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + u_N N(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) + \frac{1}{2}u_{CC}C^2\hat{c}_t + \frac{1}{2}u_{NN}N^2\hat{n}_t \text{ (reservate only to 2o)} \tag{532}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1 + \frac{u_{CC}}{u_C}C}{2}\hat{c}_t) + u_N N(\hat{n}_t + \frac{1 + \frac{u_{NN}}{u_N}N}{2}\hat{n}_t) \tag{533}$$

$$\Rightarrow \frac{u(C_t, N_t) - u(C, N)}{u_C C} = (\hat{c}_t + \frac{1 + \frac{u_{CC}}{u_C}C}{2}\hat{c}_t^2) + \frac{u_N N}{u_C C}(\hat{n}_t + \frac{1 + \frac{u_{NN}}{u_N}}{2}\hat{n}_t^2) \tag{534}$$

$$\because \hat{c}_t = \hat{y}_t \text{ (market clearing at equilibrium)} \tag{535}$$

$$u(C_t, N_t) = \frac{C_t^{1-\sigma} - 1}{1 - \sigma} - \frac{N_t^{1+\varphi} - 1}{1 + \varphi} \iff \begin{cases} -\frac{u_{CC}}{u_C}C = \sigma \\ -\frac{u_{NN}}{u_N}N = \varphi \end{cases} \tag{536}$$

Note 31 (Coefficients of CES function)**.**

σ is intertemporal substitute elasticy, φ is frisch elasticity

continue to the last taylor expanded equation, introducing var_i as the deviation of each firm's price

$$\ldots = (\hat{y}_t + \frac{1 - \sigma}{2}\hat{y}_t^2) + \frac{u_N N}{u_C C}\{\frac{1}{\alpha}(\hat{y}_t - a_t) + A \cdot var_i P_{it}\} + \frac{1 + \varphi}{2}[\frac{1}{\alpha}(\hat{y}_t - a_t) + A \cdot var_i P_{it}]^2 \tag{537}$$

$$= (\hat{y}_t + \frac{1 - \sigma}{2}\hat{y}_t^2) + \frac{u_N N}{u_C C}\frac{1}{\alpha}\{[(\hat{y}_t - a) + B \cdot var_i P_{it}] + \frac{1 + \varphi}{2\alpha}(\hat{y}_t - a_t)^2\}, B \equiv \alpha A \tag{538}$$

, here let's ignore many fluctuation terms from taylor expansion as they are extremely small and irrelevant to monetary policy.

$$-\frac{U_N}{U_C} = MP \cdot N = \alpha A N^{\alpha-1} \tag{539}$$

$$= \alpha \frac{C}{N} \Rightarrow \frac{u_N N}{u_C C} = -\alpha \text{ (can use this to simplify equation before)} \tag{540}$$

go back to before

$$\Rightarrow \ldots = (\hat{y}_t - \frac{1-\sigma}{2}\hat{y}_t^2) - \{(\hat{y}_t - a_t) + B \cdot var_i P_{it} + \frac{1+\varphi}{2\alpha}(\hat{y}_t - a_t)^2\} \tag{541}$$

$$= \frac{1-\sigma}{2}\hat{y}_t^2 - B \cdot var_i P_{it} - \frac{1+\varphi}{2\alpha}(\hat{y}_t - a_t)^2 \tag{542}$$

$$= \frac{1-\sigma}{2}\hat{y}_t^2 - B \cdot var_i P_{it} - \frac{1+\varphi}{2\alpha}(\hat{y}_t^2 - 2\hat{y}_t a_t + a_t^2) \tag{543}$$

$$= (\frac{1-\sigma}{2} + \frac{1+\varphi}{2\alpha})\hat{y}_t^2 + \frac{\sigma+\varphi}{\alpha}\hat{y}_t a_t - B \cdot var_i P_{it} \tag{544}$$

$$= \frac{\alpha(1-\alpha)+1+\varphi}{2\alpha}\hat{y}_t^2 + \frac{\sigma+\varphi}{\alpha}\hat{y}_t \hat{y}_t^f - B \cdot var_i P_{it} \text{ (remember when we work out the } \hat{y}_t^f\text{)} \tag{545}$$

$$= \underbrace{-\frac{1}{2}[\underbrace{C(\hat{y}_t - \hat{y}_t^f)}_{\text{deviation from output}} + \underbrace{D(\hat{\pi}_t - \pi^*)^2}_{\text{inflation}}]}_{\text{welfare loss}} \text{ (welfare loss function)} \tag{546}$$

the goal is to minimize the whole term

Note 32 (paradoxy in welfare loss function)**.**

the two goals are contradicated according to philips curve

18 Fiscal Policy

Planned chapter.

18.1 Government budgets, debt, and Ricardian equivalence

Derive the government flow and intertemporal budget constraints. Explain the conditions for Ricardian equivalence and why they can fail.

18.2 Taxes, purchases, transfers, and public investment

Separate income, substitution, and resource effects. Compare the objectives and transmission mechanisms of different fiscal instruments.

18.3 Fiscal multipliers and financing

Define the multiplier and its horizon. Compare spending responses across financing arrangements, monetary policy rules, household structures, and open economy environments.

19 Policy Interaction and Constraints

Planned chapter.

19.1 Fiscal and monetary regimes

Study how the fiscal backing of public liabilities interacts with interest-rate policy, inflation, and equilibrium determination.

19.2 Debt sustainability and policy space

Derive debt-to-output dynamics and distinguish arithmetic from an assessment of sustainability. Include growth, interest rates, primary balances, and risk.

19.3 Policy combinations in an open economy

Bring together exchange-rate regimes, capital flows, domestic balance sheets, and international spillovers. Compare policy combinations under stated institutional and financial constraints.

Part VI
Topics

The topics combine mechanisms from the earlier parts. Each begins with dated facts and measurement choices, compares competing explanations, and asks what evidence would distinguish them. Policy conclusions remain conditional on that evidence.

20 The Dollar in the International Monetary System

Planned chapter.

20.1 History and measures of international currency use

Separate reserve holdings, invoicing, payments, funding, and exchange-rate anchors. Define what each measure can establish about dollar dominance.

20.2 Network effects, invoicing, and safe assets

Explain the mechanisms that can sustain an international currency's use. Connect dollar pricing and demand for safe assets to the open economy models.

20.3 Benefits, obligations, and the international balance sheet

Distinguish seigniorage, financing advantages, and valuation effects. Examine the obligations and constraints associated with supplying safe and liquid international liabilities.

20.4 Global funding and policy spillovers

Trace dollar borrowing, funding stress, and the international transmission of US policy. Examine liquidity facilities and the role of financial sanctions.

20.5 Persistence, alternatives, and testable changes

Compare explanations of persistence with possible sources of change. Assess claims about de-dollarization separately across the measures defined at the start of the chapter.

21 China's Growth and Adjustment after the Pandemic

Planned chapter.

21.1 The baseline, timeline, and measurement

Separate the pre-pandemic baseline, pandemic years, and reopening period. Distinguish output levels from growth rates and real activity from nominal income. An observation after the pandemic does not identify a pandemic effect.

21.2 Demand, consumption, and employment

Compare cyclical explanations using household income, consumption, prices, and employment. State what each explanation predicts before assessing it.

21.3 Property, balance sheets, and local public finance

Map the links between housing, household and firm balance sheets, banks, and local government finances. Compare expenditure adjustment and debt resolution mechanisms.

21.4 Population, productivity, and resource allocation

Use growth accounting and the growth models to distinguish demographic, investment, and productivity explanations. Examine the allocation of resources across sectors and firms.

21.5 External demand and policy alternatives

Connect domestic adjustment to trade and international conditions. Compare policy alternatives through their mechanisms, constraints, distributional effects, and testable implications.